

RSMA for Hybrid RIS-UAV-aided Full-Duplex Communications with Finite Blocklength Codes under Imperfect SIC

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Abstract—In this work, we consider a hybrid aerial full-duplex (FD) relaying consists of a reconfigurable intelligent surface (RIS) mounted over an FD unmanned aerial vehicle (UAV) relay operating in decode and forward mode to assist the information transfer between the base station and multiple users. For better spectral efficiency, we investigate the use of rate splitting multiple access (RSMA) in such networks and focus on joint optimization of RSMA parameters, 3D-coordinates of the UAV/RIS, and phase shift matrix at the RIS along with analyzing the outage probability, block error rate (BLER) and achievable weighted sum rate for finite blocklength (FBL) and infinite blocklength (IBL) codes under imperfect successive interference cancellation (SIC) at each user and residual-self interference (RSI) at UAV. We first formulate the weighted sum rate maximization problem and adopt the block coordinate descent (BCD) method to deal with non-convex nature of the problem. Thereafter, we propose the BCD-based algorithm that jointly optimizes these parameters using a heuristic approach for optimum power allocation, a Riemannian conjugate gradient-based algorithm to get the optimal phase shift at the RIS, and an iterative algorithm to obtain the optimal UAV/RIS position. It also distributes the common rate among the users optimally. Next, with obtained optimal parameters, we further analyze the performance of the network and derive the closed-form expressions of BLER, outage probability, and average weighted sum rate. We present Monte Carlo simulation-based results to validate the accuracy of the proposed algorithms and derived expressions, and demonstrate the superiority of RSMA over non-orthogonal multiple access (NOMA) and conventional orthogonal multiple access (OMA) schemes.

Index Terms—Unmanned aerial vehicle (UAV), reconfigurable intelligent surfaces (RIS), rate splitting multiple access (RSMA), finite blocklength.

I. INTRODUCTION

WITH a never-ending demand for higher data rates and massive user/device connectivity, researchers are continuously working on developing next-generation wireless

networks such as beyond 5G (B5G) and sixth-generation (6G). The technologies like reconfigurable intelligent surface (RIS), unmanned aerial vehicle (UAV), non-orthogonal multiple access (NOMA), rate splitting multiple access (RSMA), etc., are extensively being investigated to fulfill the desired quality of services (QoSs) such as ever-increasing data rates, spectral efficiency (SE), broader coverage, etc [1], [2].

RIS uses small reflecting surfaces/elements to assist the information transfer between the communication nodes. These elements are made up of meta-material and their phase shift is controlled using software that enables smart reflection-aided transmission [3]–[5]. RIS comprises reconfigurable reflective surfaces that employ tunable subwavelength structures (e.g., microstrip patches) to modulate the phase and/or amplitude of reflected waves [6]. It can be noted that the RIS is neither part of the transmitter radio nor the receiver radio. It is a novel way to control the wireless propagation medium, which until now had been deemed as uncontrollable. Accordingly, the fundamental role of a RIS is to affect the dispersion of wireless signals transmitted by other devices, without producing its signals. Therefore, high efficient energy, better spectrum utilization, low implementation cost, and flexible deployment of RIS are some of the major advantages that lead to the huge popularity of RIS-assisted communication. In recent literature, the application of RIS in wireless communication networks has been well investigated by researchers [7]–[11]. For example, in [8], the performance of a RIS-assisted point-to-point communication was investigated where the base station (BS) communicated the information symbols to ground users (GU) using a RIS in the absence of a direct link. In [9], a concept of relaying assisted communication was introduced along with RIS, where its controller also acts as a decode and forward (DF) relay to assist the information exchange between the BS and a GU. The achievable rate was maximized by jointly optimizing the passive beamforming phase of the RIS elements along with time allocations for controller-based DF relay. Further, the impact of resource allocation optimization on the achievable capacity was studied in [10] for a RIS-aided multiuser communication system.

A. Related Works

Recently, inspired by the concept of the aerial relay, researchers proposed and studied the idea of aerial RIS (ARIS) for wireless communication networks in which a RIS is

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mounted over an aerial platform such as an air balloon or UAV [12]. Along with easy and quick deployment, this provides an additional advantage or degree of freedom to RIS where it can adjust its location according to the position of ground devices (BS or GUs) to attain a line of sight (LoS) channels and improve the network performance [13]–[15]. In [13], the performance of ARIS was studied to maximize the worst-case signal-to-noise ratio (SNR) at a single GU as compared to conventional/fixed RIS. The major reasons for this improvement are better deployment capability, more appropriate height to achieve LoS path, and wider signal-view reflection. The use of multiple ARIS, each mounted over UAV, and a fixed RIS for simultaneous information and wireless power transfer (SWIPT) was demonstrated in [14], where the achievable throughput was maximized by jointly optimizing the power-splitting (PS) ratio and phase shift of all the RIS elements. *The aforementioned works [14], [15] used UAV only as a carrier and did not explore its other features such as a relay [16]–[18] making them less efficient in terms of resource utilization.* In short, because of low cost, quick deployment, easily controllable mobility, easy implementation, etc., using UAV as a full duplex (FD) relay provides much better performance compared to conventional relaying [18].

In order to provide more efficient use of UAV-aided ARIS, motivated by the above, a novel idea of using hybrid RIS-FD-UAV relay in a single user downlink network was studied in [19] wherein the height and number of elements were optimized separately for maximizing the energy efficiency of the network. Additionally, it used a conventional infinite blocklength (IBL) messaging scheme with an asymptotically large blocklength for the investigation of RIS-UAV communication. However, to support reliable and low latency communication in 6G and Industry 4.0, short packet transmission has been well-motivated in such networks that use a finite blocklength (FBL) messaging scheme to obtain desired QoS [20]. Note that due to channel dispersion, net performance in FBL is reduced as compared to IBL and the results for IBL cannot directly be used to analyze/study the performance of the FBL scheme. Therefore, it is interesting and extremely important to investigate the performance of hybrid RIS-FD-UAV using the FBL scheme in short packet multiuser communication networks.

Further, in literature, there are several multiple access techniques such as NOMA, RSMA, etc., that are used to provide simultaneous information transfer in a multiuser communication. Both NOMA and RSMA efficiently exploit the available spectrum by allowing multiple users to share the same resource (time/frequency/code) using power domain multiplexing at the transmitter and successive interference cancellation (SIC) at the receiver [21]. However, unlike the use of multi-stage SIC at NOMA which has computational complexity directly proportional to the number of users, RSMA uses two-step decoding (including single-stage SIC) at each receiver, resulting in enhanced performance at reduced complexity compared to NOMA [22]. The dominance of RSMA on the performance of wireless networks over other multiple access techniques such as NOMA, orthogonal multiple access (OMA), etc., was demonstrated in numerous studies [20], [23], [24]. Besides,

NOMA can always be obtained as a special case of RSMA [25]. Therefore, through this work, we introduce the use of RSMA in a hybrid RIS-FD-UAV-aided multiuser network. In particular, each user's message is divided into common and private messages at the transmitters' side using one-layer rate splitting (RS). Then, a single common stream is formed by combining the common messages using a predefined codebook. This common stream/message is decoded by each user whereas a private message is decoded only by the intended user using one-step SIC [24]. Moreover, in literature, the performance of RSMA was studied separately for both RIS and UAV-aided communication networks [26]–[29]. In [27], the impact of RSMA on the outage probability and throughput of a UAV-aided multiuser downlink network was investigated. The leverage between RSMA and RIS was presented in [29] and outage probabilities of the users were derived using on-off control technique-based phase shift matrix.

B. Motivation and Contribution

Consequently, the integration of RIS and FD-UAV relaying with some of the most advanced next-generation techniques such as RSMA and FD communication can further improve the performance of the wireless networks as compared to conventional/fixed RIS or only FD-UAV-aided transmission through efficient resource utilization. In practical scenarios, like disaster management in high terrains and extreme environmental conditions, this type of relaying can be very effective in providing end-to-end communication. Sometimes RIS suffers from excessive multiplicative fading and cannot provide desired QoS, in such scenarios the FD-UAV transmission can be extremely useful¹. In addition to this, to achieve the futuristic demands of modern generation wireless networks the inclusion of ultra-reliable low latency communication in R-15 of 5G standards [30] has gained the research attention towards the utilization of FBL-codes. In this context, in [19], the integration of UAV and RIS has been studied by considering the fixed UAV/RIS position and ideal phase shift matrix at RIS. However, these assumptions are unrealistic for practical applications. Therefore, to realize the full potential of these technologies in view of advanced wireless applications, it is essential to explore the capabilities of hybrid RIS-FD-UAV-assisted communication considering practical assumptions such as variable phase shift matrix of RIS, optimal position of UAV, and low latency codes. Further, to mitigate the SIC error propagation in multiuser scenario RSMA² can be preferred over NOMA. Therefore, in this work, we investigate the performance of a multiuser wireless communication network consisting of a BS, multiple GUs, and a hybrid RIS-FD-UAV that assists the information

¹Since, RIS provides an inherent FD ability. In contrast to this, the half-duplex (HD) operation at UAV can lower the spectral efficiency by half and the performance of RIS-aided transmission will always be superior in comparison to HD-UAV-based transmission. Sometimes when RIS fails due to excessive multiplicative fading loss then the HD UAV mode cannot provide efficient information exchange. Therefore, in the UAV-aided transmission, the FD operation is required for information exchange between BS and GUs.

²Note that RSMA can get similar performance to NOMA but with a single SIC, therefore significantly reducing the receiver complexity [24]

transfer from BS to all GUs using RSMA signaling. The major contributions of this work are as follows:

- For the first time, we investigate RSMA for a hybrid RIS-FD-UAV-aided multi-user wireless communication network and jointly optimize the 3D-coordinates of the UAV/RIS and phase shift matrix at the RIS and analyze the outage probability, BLER, and achievable weighted sum rate for FBL and IBL schemes under the practical consideration of imperfect SIC at each user and residual self-interference (RSI) at UAV³.
- Unlike [19] wherein only the height of the UAV is optimized for a single user considering the idealized phase shift matrix, we formulate an optimization problem that jointly optimizes RSMA parameters, the phase shift matrix and 3D-position of the UAV/RIS to maximize the achievable weighted sum rate under the imperfect SIC and RSI. Due to the non-convex nature of the formulated problem, we first reformulate the problem into a simpler form by replacing the summation of the weighted distributed common rate with its maximum value.
- Thereafter, we utilize the block coordinate descent (BCD) method to convert the formulated optimization problem into three blocks 1) power allocation for fixed phase shift matrix and position 2) phase shift matrix optimization for fixed power allocation and position, and 3) position optimization for fixed power allocation and phase shift matrix. The optimum power allocation coefficients are obtained using a heuristic approach. Thereafter, a Riemannian conjugate gradient (RCG) method based algorithm is proposed to determine the optimal phase shift at the RIS. Next, an iterative algorithm is proposed to obtain the optimal RIS/UAV coordinates.
- Consequently, we propose a BCD-based algorithm that jointly optimizes power allocation, phase shift matrix, and UAV/RIS position using the combination of aforementioned sub-methods until the convergence is achieved. In addition to this, it uses the optimal values of these parameters to obtain the optimum common rate for each user.
- We further analyze the network performance with the obtained optimal phase shift matrix at the RIS and UAV/RIS position. In this regard, to tackle the analytical complexity due to the involvement of RIS-aided channel, imperfect SIC, and RSI, we use the properties of the series of Laguerre polynomials and standard mathematical tools such as Riemann integral approximation, Gaussian-Laguerre quadrature, etc., and derive the closed-form expressions for the user BLER, outage probabilities, and average weighted sum rate.
- We validate the accuracy of the proposed algorithm and derived analytical expressions through extensive simulations, and illustrate the advantages of the hybrid RIS-FD-UAV-aided mode under both IBL and FBL codes.

³Note that due to RSI and imperfect SIC, the analytical complexity of signal-to-interference-plus-noise-ratio (SINR) at UAV and each user increases considerably. Therefore, compared to perfect SIC and no RSI case, the analysis of the network with imperfect SIC and RSI becomes very difficult and extremely challenging.

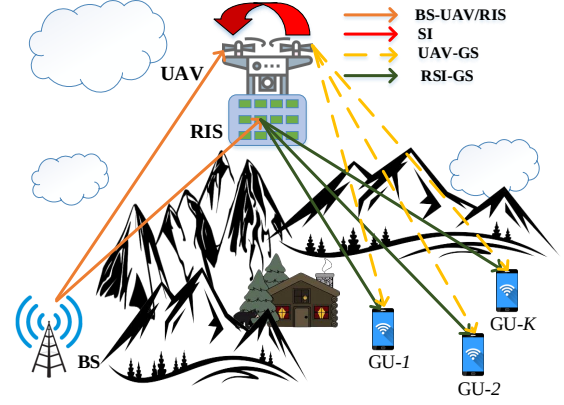


Fig. 1: Hybrid RIS-FD-UAV-aided Communication network.

In particular, when RIS has a lower number of reflecting elements, UAV-only mode outperforms the RIS-only mode, however, with higher number of reflecting elements and low available transmit power at UAV, the RIS-only mode dominates the UAV-only mode. Thus, due to selection combining (SC), hybrid RIS-FD-UAV-aided mode always gives the best performance as compared to other two modes for both IBL and FBL codes. In addition to this, we show that, with an imperfection of around 5%–10% in SIC, RSMA provides approximately 7%–16% higher average weighted sum rate compared to NOMA and conventional OMA schemes, which confirms the superiority of RSMA over NOMA and OMA. Finally, we highlight the impact of the number of users on the average BLER performance of the considered network.

C. Organization of the Paper

Section II discusses the system model. Section III presents the joint phase shift and position of the UAV/RIS optimization problem and its solution. Closed-form expressions of the performance metrics are derived in Section IV and Section V for FBL and IBL, respectively, schemes. Section VI discusses the obtained results while Section VII concludes the paper.

II. SYSTEM MODEL

We consider a hybrid aerial FD communication network consisting of a UAV-mounted RIS, as shown in Fig. 1. The RIS consists of N reflecting elements to aid the communication between the BS (b) to K GUs. The BS is equipped with a single transmit antenna while each GU consists of a single receive antenna. In addition to this, the UAV (u) also acts as an independent FD-DF relay and has separate transmit and receive antennas to enable simultaneous transmission and reception. Further, important symbols used throughout this paper are summarized in Table I.

A. Aerial Pathloss Model

UAV-assisted ARIS-aided communication generally follows the aerial channel model that considers the impact of small-scale as well as large-scale fading on the net channel gain [19].

TABLE I: Important Symbols

Symbol	Parameter	Symbol	Parameter
$\mathcal{I} = \{I_1, I_2, \dots, I_N\}$	Set of N RIS elements	$\mathbf{b}$ and e	Rician factor and Euler's number
$\mathcal{K} = \{1, 2, \dots, K\}$	Set of K GUs	$\Xi(\cdot)$	Error in SIC
$\{x_i, y_i, z_i\}$	Coordinates of device- $i \in \{u, \mathcal{I}, b, \mathcal{K}\}$	P_b & P_u	Transmit power at BS & UAV
$g(\cdot), \beta(\cdot)$ & $\hat{g}(\cdot)$	Channel gain, large & small scale fading	$P_{O(\cdot)}, \xi(\cdot)$ & $R(\cdot)$	Outage probability, BLER, & rate
$\{\psi(\cdot), \psi(\cdot), \mathbf{q}(\cdot), \mathbf{u}(\cdot)\}$	Environmental constants [19]	$Q(\cdot, \cdot)$ & $Q(\cdot)$	Marcum & Gaussian Q -function
$\mathcal{CN}(\cdot, \cdot), \mathbb{E}(\cdot)$ & $\Re(\cdot)$	Complex Gaussian, expectation and real value	Θ	Diagonal phase-shift matrix at RIS
$\phi_n \forall n \in \mathcal{I}$	Phase shift at each element	$S(\cdot), f(\cdot)(\cdot)$ & $\mathbb{F}(\cdot)(\cdot)$	SINR, PDF & CDF
Δ and grad	Euclidean and Riemannian gradient	$\mathcal{O}$	Big O notation
$\Gamma(\cdot)$ & $\Gamma(\cdot, \cdot)$	Gamma & Incomplete Gamma function	$s_{(\cdot)}^{th}$ & $\hat{s}_{(\cdot)}^{th}$	SINR threshold
$K(\cdot)$ and $L(\cdot)$	Modified Bessel's function & Laguerre polynomial	$r(\cdot)$	Blocklength

In particular, the channel gain for the $i-j$ ($i, j \in \{u, \mathcal{I}, b, \mathcal{K}\}$) link is effected by both large scale fading (β_{ij}) and small scale fading ($\hat{g}_{ij}$) and can be expressed as

$$g_{ij} = \sqrt{\beta_{ij}} \hat{g}_{ij}. \quad (1)$$

Here, $\hat{g}_{ij}$ represents follows Rician distribution [19] and is given by

$$\hat{g}_{ij} = \left(\sqrt{\frac{\mathbf{b}}{\mathbf{b}+1}} + \sqrt{\frac{1}{\mathbf{b}+1}} h_{ij} \right), \quad (2)$$

where $\mathbf{b}$ denotes the Rician factor and $h_{ij} \sim \mathcal{CN}(0, 1)$. Note that when $\mathbf{b} \rightarrow 0$, $\hat{g}_{ij} \rightarrow h_{ij}$ [31]. Further, β_{ij} in (1) is given by

$$\beta_{ij} = \beta_o d_{ij}^{-\alpha(\Lambda_{ij})}, \quad (3)$$

where β_o denotes the channel gain at the reference distance of 1 m, and d_{ij} represents the distance between devices i and j which is given by

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \quad (4)$$

$\alpha(\Lambda_{ij})$ in (3) denotes the position-dependent pathloss exponent which can be evaluated as

$$\alpha(\Lambda_{ij}) = \Pr(\Lambda_{ij}) \mathbf{q}_{ij} + \mathbf{u}_{ij}, \quad (5)$$

where $\Pr(\Lambda_{ij}) = (1 + \psi_{ij} \exp(\hat{\psi}_{ij}(\Lambda_{ij} - \psi_{ij})))^{-1}$, $\Lambda_{ij} = \arctan\left(\frac{z_i - z_j}{\sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}}\right)$, $\mathbf{q}_{ij}$, $\mathbf{u}_{ij}$, ψ_{ij} , and $\hat{\psi}_{ij}$ are constants given in [19], [32]. Unlike constant pathloss exponent which is independent of the position/location of the UAV [33], (5) provides a more realistic 3D-position dependent and an average pathloss exponent approximated over a reasonable range of communication for a UAV [34].

B. Information Transmission

Since RSMA signalling is used, thus, the message of each user is divided into two parts: common part ($W_{k,c}$) and private part ($W_{k,p}$). Next, the common part of all users is combined and encoded as a common message (s_c) using a common codebook available on each device. Further, the private part ($W_{k,p}$) of GU- k is encoded as private message (s_k) [24]. Note that due to this encoding process, similar to [24], the correlation among the messages is assumed to zero i.e. $\mathbb{E}[s_i s_j] = 1$ and $\mathbb{E}[s_i s_j] = 0 \forall i \neq j, i, j \in \{c, \mathcal{K}\}$. s_c will be decoded by each user whereas s_k will be decoded

at GU- k only treating all other $s_j \forall j \neq k$ as interference. The superimposed symbol s which is obtained by combining common message and private messages of all users as

$$s = \sqrt{\alpha_c P_b} s_c + \sum_{k \in \mathcal{K}} \sqrt{\alpha_k P_b} s_k, \quad (6)$$

where P_b denotes the transmit power of BS, and α_c and α_k are the power coefficients allocated to s_c and s_k , respectively, with $\alpha_c + \sum_{k \in \mathcal{K}} \alpha_k = 1$. Further, based on the operation of RIS and UAV, the information transmission between BS to GUs occurs in following transmission modes:

1) *RIS-aided Transmission*: In this mode, RIS assists the information transfer from BS to UAV using N reflecting elements. BS transmits the superimposed symbol s to RIS and each element reflects the respective signal to all users with a phase shift of $\phi_n \forall n \in \mathcal{I}$. The overall signal received at k^{th} user is a combination of signals from N reflecting elements and given by

$$\begin{aligned} Y_{kI} &= \mathbf{g}_{kI}^H \Theta \mathbf{g}_{Ib} s + w_k \\ &= \underbrace{\sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n \sqrt{\alpha_c P_b} s_c}_{\text{common message}} + \underbrace{\sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n \sqrt{\alpha_k P_b} s_k}_{\text{private message}} \\ &\quad + \underbrace{\sum_{i \in \mathcal{K}, i \neq k} \sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n \sqrt{\alpha_i P_b} s_i}_{\text{Interference}} + \underbrace{w_k}_{\text{AWGN}}, \end{aligned} \quad (7)$$

where $\mathbf{g}_{kI} = [g_{kI_1}, g_{kI_2}, \dots, g_{kI_N}]^H$, $\mathbf{g}_{Ib} = [g_{I_1b}, g_{I_2b}, \dots, g_{I_Nb}]^H$, w_k is the AWGN with variance σ^2 , and $\Theta = \text{diag}(\boldsymbol{\theta})$ represents the diagonal phase-shift matrix at RIS with $\boldsymbol{\theta} = [\theta_{I_1}, \theta_{I_2}, \dots, \theta_{I_N}]^H$ and $\theta_n = e^{j\phi_n}$. Using (1), (6) and (7), the signal-to-interference-plus-noise-ratio (SINR) for decoding the common message at k^{th} GU is given by

$$S_{k,cI} = \frac{|\sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n|^2 \alpha_c P_b}{|\sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n|^2 P_b (\sum_{i \in \mathcal{K}} \alpha_i) + \sigma^2}. \quad (8)$$

After performing SIC, the SINR to decode its private message at the k^{th} GU in RIS-aided mode can be expressed as

$$S_{kI} = \frac{|\sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n|^2 \alpha_k P_b}{|\sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n|^2 P_b (\sum_{i \in \mathcal{K}, i \neq k} \alpha_i + \Xi_{k,cI} \alpha_c) + \sigma^2}, \quad (9)$$

where $\Xi_{k,cI} \in [0, 1]$ denotes the error in SIC⁴.

⁴Due to decoding errors, a slight misdetection of the common message may occur. Unlike a perfect SIC case, this misdetection might lead to an error in SIC and there will be an interference which will be a fraction of the net received power corresponding to a common message [28].

2) *UAV-aided Transmission*: In this mode, the UAV acts as an FD DF⁵ relay to assist the information transfer from BS to GUs. The sampled matched filter output at UAV is given by

$$Y_u = g_{ub}s + \sqrt{\vartheta}g_{uu}\hat{s} + w_u$$

$$= \underbrace{\sqrt{\alpha_c P_b} g_{ub} s_c}_{\text{common message}} + \underbrace{\sum_{i \in \mathcal{K}} \sqrt{\alpha_i P_b} g_{ub} s_i}_{\text{private messages}} + \underbrace{\sqrt{\vartheta} g_{uu} \hat{s}}_{\text{RSI}} + \underbrace{w_u}_{\text{AWGN}}, \quad (10)$$

where $\sqrt{\vartheta}g_{uu}\hat{s}$ represents the residual self-interference (RSI) with $\vartheta \in [0, 1]$ denoting the RSI cancellation factor, $g_{uu} \sim \mathcal{CN}(0, 1/\lambda_{uu})$ denotes the channel between transmit and receive antenna of the UAV with zero mean and λ_{uu} variance, w_u is the AWGN at UAV, and $\hat{s}$ denotes the re-encoded symbol at UAV which can be expressed as

$$\hat{s} = \sqrt{\alpha_c P_u} \hat{s}_c + \sum_{k=1}^K \sqrt{\alpha_k P_u} \hat{s}_k, \quad (11)$$

where $\hat{s}_k$ represents the symbol of the k^{th} GU at UAV, $\hat{s}_c$ denotes the encoded common message, and P_u is the available transmit power at the UAV. Using (1) and (10), SINR to decode s_c at UAV is given by

$$S_{cu} = \frac{|g_{ub}|^2 \alpha_c P_b}{|g_{ub}|^2 P_b (\sum_{k \in \mathcal{K}} \alpha_k) + P_{SI} + \sigma^2}, \quad (12)$$

where $P_{SI} = \vartheta |\lambda_{uu}|^2 P_u$ represents the power of RSI⁶ signal. After decoding s_c , UAV applies multi-step SIC and decodes the private messages of all users. Thus, the SINR to decode the private message of GU- k at UAV is given by

$$S_{ku} = \frac{|g_{ub}|^2 \alpha_k P_b}{\left\{ |g_{ub}|^2 P_b (\sum_{i \in \mathcal{K}_l^k} \alpha_i + \sum_{j \in \mathcal{K}_h^k} \Xi_{ju} \alpha_j) + |g_{ub}|^2 P_b \Xi_{cu} \alpha_c + P_{SI} + \sigma^2 \right\}}, \quad (13)$$

where Ξ_{cu} and $\Xi_{ju} \in [0, 1]^7$ denote the SIC error, $\mathcal{K}_l^k$ and $\mathcal{K}_h^k$ represent the set of all users having lower and higher power coefficients, respectively, than α_k . Next, similar to BS, the UAV uses RSMA signalling to transmit the superimposed symbol $\hat{s}$ to all the users. The received signal at each GU is given by

$$Y_{ku} = g_{ku}\hat{s} + w_k = \underbrace{g_{ku}\sqrt{\alpha_c P_u}\hat{s}_c}_{\text{common message}} + \underbrace{g_{ku}\sqrt{\alpha_k P_u}\hat{s}_k}_{\text{private message}} + \underbrace{g_{ku} \sum_{i \in \mathcal{K}, i \neq k} \sqrt{\alpha_i P_u} \hat{s}_i}_{\text{interference}} + \underbrace{w_k}_{\text{AWGN}}, \quad (14)$$

where w_k denotes the AWGN with variance σ^2 . Using (1) and (14), the SINR for decoding the common message at GU- k is given by

$$\hat{S}_{k,cu} = \frac{|g_{ku}|^2 \alpha_c P_u}{|g_{ku}|^2 P_u (\sum_{i \in \mathcal{K}} \alpha_i) + \sigma^2}. \quad (15)$$

⁵Although the AF requires simple circuitry than the DF, the noise amplification limits the performance of the AF relaying. Therefore, DF protocol is preferred over AF for relaying to ensure a better quality of service in cooperative systems [35].

⁶Note that using active and passive SI cancellation techniques [36], high quality SI suppression can be achieved. Therefore, considering the fact that $|g_{uu}|^2$ is very small, similar to [37], we assume $|g_{uu}|^2 \approx \lambda_{uu}$.

⁷Note that decoding of common and each user's private message at UAV is performed using multi-step SIC.

After performing SIC, the SINR to decode the desired private message at GU- k is given by

$$\hat{S}_{ku} = \frac{|g_{ku}|^2 \alpha_k \beta_{ku} P_u}{|g_{ku}|^2 P_u (\sum_{i \in \mathcal{K}, i \neq k} \alpha_i + \Xi_{k,cu} \alpha_c) + \sigma^2}, \quad (16)$$

where $\Xi_{k,cu} \in [0, 1]$ denotes to the error in SIC.

3) *Hybrid RIS-FD-UAV-aided Transmission Mode*: In this transmission mode, both UAV and RIS assist the information exchange from BS to GU. In particular, BS transmits the superimposed signal to both UAV and RIS. Further, UAV decodes and forwards the re-encoded symbols to each user using RSMA signalling. At the same time, RIS reflects the incoming symbols to each user with a certain phase shift. Since each user will receive signals from both UAV and RIS, therefore, uses SINR-based selection combining (SC)⁸ to decode the desired symbols.

III. JOINT PHASE SHIFT AND RIS/UAV-PLACEMENT OPTIMIZATION

This section discusses the joint optimization of the phase shift matrix of RIS along with the position of RIS/UAV to maximize the network's performance in terms of the weighted sum rate for each mode with FBL transmission. BS transmits the short packet with a fixed blocklength to each user formed by combining the common and private messages (see (6)). Next, we provide a detailed framework to maximize the weighted sum rate for each transmission mode.

1) *RIS-aided Transmission*: Using (8), the achievable rate at k^{th} user while decoding common message for FBL messaging with a blocklength of r_c can be expressed as [38]

$$R_{k,cI}^{FBL} \approx \log_2(1 + S_{k,cI}) - Q^{-1}(\tilde{\xi}_{kc}) \log_2 e \sqrt{(1 - (1 + S_{k,cI})^{-2}) / r_c}, \quad (17)$$

where $\tilde{\xi}_{kc}$ is the constant tolerable/desired BLER at k^{th} user for decoding common message. Also, $1 - (1 + S_{k,cI})^{-2}$ constitutes to the channel dispersion. Similarly, the achievable rate at k^{th} user while decoding a private message can be expressed as

$$R_{k,pI}^{FBL} \approx \log_2(1 + S_{kI}) - Q^{-1}(\tilde{\xi}_{kp}) \log_2 e \sqrt{(1 - (1 + S_{kI})^{-2}) / r_{kp}}, \quad (18)$$

where $\tilde{\xi}_{kp}$ is the constant tolerable BLER at k^{th} user for decoding private message having a blocklength of r_{kp} . According to the principle of RSMA [39], the total weighted sum rate of the considered system can be expressed as

$$R_{I,\text{sum}}^{FBL} = \sum_{k \in \mathcal{K}} \eta_k (R_{kc}^{FBL} + R_{k,pI}^{FBL}), \quad (19)$$

⁸In the case of SC, the combiner outputs the signal on the branch with the highest SNR, which requires one receiver to switch to the active branch, and co-phasing of multiple branches is not required as is the case with the other combining techniques such as MRC [19]. Therefore, SC exhibits low overhead and provides analytical simplicity. Also, SC has the simplest receiver implementation since the RIS-only and UAV-only transmission links are independent, therefore, to reduce the receiver complexity SC is performed at the destination receiver.

where η_k is the weight and R_{kc}^{FBL} is the common rate allocated to k^{th} user satisfying $\sum_{k \in \mathcal{K}} \eta_k R_{kc}^{FBL} \leq \min_{k \in \mathcal{K}} R_{k,cI}^{FBL}$. The optimization problem to maximize the ergodic sum rate of the considered network can be formulated as

$$(P0) \quad \max_{\substack{\theta, \tilde{\xi}_{kc}, \tilde{\xi}_{kp}, x_u, y_u, z_u, \\ \alpha_c, \alpha_k, R_{kc}^{FBL} \forall k \in \mathcal{K}}} R_{I,\text{sum}}^{FBL} = \sum_{k \in \mathcal{K}} \eta_k (R_{kc}^{FBL} + R_{k,pI}^{FBL}) \quad (20)$$

$$\text{s.t. } |\theta_n| = 1, \forall n \in \mathcal{I}, \quad (21)$$

$$\tilde{\xi}_{kc} \leq \tilde{\xi}_{kc}^{\max}, \tilde{\xi}_{kp} \leq \tilde{\xi}_{kp}^{\max} \quad \forall k \in \mathcal{K} \quad (22)$$

$$\alpha_c + \sum_{k \in \mathcal{K}} \alpha_k = 1 \quad (23)$$

$$\sum_{k \in \mathcal{K}} \eta_k R_{kc}^{FBL} \leq \min_{k \in \mathcal{K}} R_{k,cI}^{FBL} \quad (24)$$

$$R_{kc}^{FBL} + R_{k,pI}^{FBL} \geq R_{k,\min} \quad \forall k \in \mathcal{K} \quad (25)$$

$$x_{\min} \leq x_u \leq x_{\max}, y_{\min} \leq y_u \leq y_{\max}, \quad (26)$$

$$z_{\min} \leq z_u \leq z_{\max},$$

where $(\cdot)_{\max}$ and $(\cdot)_{\min}$ denote the maximum and minimum values of the respective parameters, (22) corresponds to the maximum tolerable BLER constraints ($\tilde{\xi}_{kc}^{\max}$ and $\tilde{\xi}_{kp}^{\max}$) for the respective messages at each user, (23) and (24) are the allocated power coefficients and achievable common rate constraints, the constraint (25) ensures a minimum rate $R_{k,\min}$ at each user, and (26) corresponds to the flying limitations of the UAV. Note that due to the coupling of variables and involvement of their ratio in the objective function (OF), problem (P0) is non-convex in nature. Therefore, it is extremely difficult to find a solution for problem (P0). Thus, considering the fact that $\tilde{\xi}_{kc}$ and $\tilde{\xi}_{kp}$ are independent, we first obtain their optimal value as follows:

Lemma 1. *The $R_{I,\text{sum}}^{FBL}$ will be maximum w.r.t. $\tilde{\xi}_{kc}$ and $\tilde{\xi}_{kp}$ at the optimal value $\tilde{\xi}_{kc}^o = \tilde{\xi}_{kc}^{\max}$ and $\tilde{\xi}_{kp}^o = \tilde{\xi}_{kp}^{\max}$, respectively, for all $k \in \mathcal{K}$.*

Proof. To prove this lemma, we use the contradiction method and assume that the optimal solutions $\tilde{\xi}_{kc}^o$ and $\tilde{\xi}_{kp}^o \forall k \in \mathcal{K}$ satisfy strict inequality $\tilde{\xi}_{kc} < \tilde{\xi}_{kc}^{\max}$ and $\tilde{\xi}_{kp} < \tilde{\xi}_{kp}^{\max}$, respectively. With this assumption, in problem (P0), the values of $\tilde{\xi}_{kc}^o$ and $\tilde{\xi}_{kp}^o$ can be increased to achieve a higher sum rate while satisfying constraint (22), hence, contradicting their optimality.

Alternatively, it can be seen that in order to maximize $R_{I,\text{sum}}^{FBL}$, the negative term $(Q^{-1}(\tilde{\xi}_{kc}) \log_2 e \sqrt{(1 - (1 + S_{k,cI})^{-2})/r_c})$ in (17) need to be minimized w.r.t. $\tilde{\xi}_{kc}$. This can be achieved by minimizing $Q^{-1}(\tilde{\xi}_{kc})$ which in turn can be achieved by maximizing $\tilde{\xi}_{kc}$, thus we have $\tilde{\xi}_{kc}^o = \tilde{\xi}_{kc}^{\max}$. Similarly, we have $\tilde{\xi}_{kp}^o = \tilde{\xi}_{kp}^{\max}$. ■

It is note worthy that, similar to (22), the objective function (OF) of the problem (P0) will be maximum when $\sum_{k \in \mathcal{K}} \eta_k R_{kc}^{FBL} = \min_{k \in \mathcal{K}} R_{k,cI}^{FBL}$ in constraint (24). Further, for simplicity, in order to make problem (P0) feasible, we assume that P_b is sufficiently high such that there exists at least a possible combination of the variables of the problem

(P0) that satisfies rate constraint in (25). Thus, the problem (P0) can be reformulated as

$$(P0.1) \quad \max_{\substack{\theta, x_u, y_u, z_u, \\ \alpha_c, \alpha_k \forall k \in \mathcal{K}}} R_{I,\text{sum}}^{FBL} = \min_{k \in \mathcal{K}} R_{k,cI}^{FBL} + \sum_{k \in \mathcal{K}} \eta_k R_{k,pI}^{FBL} \quad (21), (23) \text{ and } (26) \text{ of } (P0).$$

Note that, due to the involvement of channel dispersion and other constraints, problem (P0) is still non-convex for obtained $\tilde{\xi}_{kc}^o$ and $\tilde{\xi}_{kp}^o \forall k \in \mathcal{K}$. Therefore, to further simplify the problem, we use approximation $(1 - (1/(1 + S_{k,cI})^2)) \approx 1$ for $S_{k,cI} \gg 1$. Thus, assuming $\hat{k} = \arg \min_{k \in \mathcal{K}} R_{k,cI}^{FBL}$, the total channel dispersion can be expressed as $\mathcal{D} = Q^{-1}(\tilde{\xi}_{\hat{k}c}^{\max}) \log_2 e / \sqrt{r_c} + \sum_{k \in \mathcal{K}} \eta_k Q^{-1}(\tilde{\xi}_{\hat{k}p}^{\max}) \log_2 e / \sqrt{r_{kp}}$. Note that $\mathcal{D}$ is constant w.r.t. all the optimization variables in problem (P0.1). Therefore, it can be reformulated as

$$(P1) \quad \max_{\substack{\theta, x_u, y_u, z_u, \\ \alpha_c, \alpha_k \forall k \in \mathcal{K}}} \hat{R}_{I,\text{sum}}^{FBL} \quad (28)$$

s.t. (21), (23) and (26) of (P0),

where, $\hat{R}_{I,\text{sum}}^{FBL}$ is obtained using (8) and (9), and after some mathematical simplification as

$$\begin{aligned} \hat{R}_{I,\text{sum}}^{FBL} = & \log_2 \left(\frac{|\theta^H \mathbf{a}_{\hat{k},c}|^2}{\sum_{i \in \mathcal{K}} |\theta^H \mathbf{a}_{\hat{k},i}|^2 + \sigma^2} \right) \\ & + \sum_{k \in \mathcal{K}} \eta_k \log_2 \left(\frac{|\theta^H \mathbf{a}_{k,k}|^2}{\sum_{i \in \mathcal{K}, i \neq k} |\theta^H \mathbf{a}_{k,i}|^2 + \Xi_{k,cI} |\theta^H \mathbf{a}_{k,c}|^2 + \sigma^2} \right) \end{aligned} \quad (29)$$

where $\mathbf{a}_{l,j} = \text{diag}(\mathbf{g}_{lI}^H) \mathbf{g}_{lB} \sqrt{P_b \alpha_j} (\forall l \in \{\hat{k}, k\}, j \in \{c, K\})$.

Next, we adopt the BCD method which divides the main problem into sub-problems (blocks) and maximizes the OF sequentially w.r.t. parameter/coordinate of each block with fixed coordinates of all other blocks to solve this problem. Therefore, using BCD, the problem (P1) can be divided into three blocks: 1) to allocate the sub-optimum power coefficients for fixed phase shift matrix and position, 2) to find the sub-optimal phase shift matrix for the fixed position and power coefficients, and 3) to find a sub-optimal position for the fixed phase shift matrix and power coefficients. Specifically, we first fetch sub-optimal values of the power coefficients $\{\alpha_c, \alpha_1, \alpha_2, \dots, \alpha_K\}$ for fixed $\{x_u, y_u, z_u\}$ and θ . These values are obtained from a sample space $\mathcal{P}$ which is a matrix of size $V \times (K + 1)$ with V representing the total possible combination of values of power allocation coefficients that satisfy (23) i.e. summation of each row is equal to one. n^{th} column of $\mathcal{P}$ represents the values of α_{n-1} except the first column which represents the values of α_c .

Sub-Optimal Phase shift matrix for fixed UAV/RIS position: For fixed position of UAV/RIS, problem (P0) can be simplified in a sub-optimal problem for maximizing the weighted sum rate w.r.t. phase shift as follows:

$$(P2) \quad \max_{\theta} \hat{R}_{I,\text{sum}}^{FBL} \quad (30)$$

s.t. (21) of (P0),

Clearly, the complexity of problem increases with increase in number of elements at RIS. However, it can be observed that $\hat{R}_{I,\text{sum}}^{FBL}$ is a continuous and differentiable function w.r.t. θ , and (21) forms a complex circle manifold [40]. Therefore, RCG method can be used to find a probable solution to problem (P2). RCG is a widely used method for designing different parameters such as precoder, phase shift matrix, etc., in the case of RIS-aided communication [40]. Note that, similar to [41], one can view the constraints of the problem (P2) as a Riemannian manifold. In particular, the rich geometry of Riemannian manifolds makes it possible to define gradients of cost functions. More importantly, the optimization over a manifold is locally analogous to that in Euclidean space. Therefore, optimization techniques that were developed for Euclidean space, e.g., gradient descent and trust-region methods have counterparts on manifolds. Since the objective function of the problem (P2) is continuous and differentiable, one straightforward approach is to use the projected gradient method. But instead of taking the Euclidean gradient, a better approach is to use the Riemannian gradient, which is formed by projecting the Euclidean gradient onto the tangent space of the complex circle manifold formed by the unit modulus constraints [42]. The RCG further speeds up the convergence by performing a gradient-dependent line search on the Riemannian manifold rather than the Euclidean space [42]. In RCG based algorithm, the following three steps are followed in each iteration:

Riemannian Gradient: The Euclidean gradient of $\hat{R}_{I,\text{sum}}^{FBL}$ is evaluated as

$$\Delta \hat{R}_{I,\text{sum}}^{FBL}(l+1) = 2(\mathbf{A}_{\hat{k}_c} + \sum_{k=1}^K \eta_k \mathbf{A}_{k_p}), \quad (31)$$

where $\mathbf{A}_{\hat{k}_c} = \frac{(\mathbf{a}_{\hat{k},c} \mathbf{a}_{\hat{k},c}^H \theta)}{|\theta^H \mathbf{a}_{\hat{k},c}|^2} - \frac{\sum_{i \in \{K\}} (\mathbf{a}_{\hat{k},i} \mathbf{a}_{\hat{k},i}^H \theta)}{\sum_{i \in \{K\}} |\theta^H \mathbf{a}_{\hat{k},i}|^2 + \sigma^2}$ and $\mathbf{A}_{k_p} = \frac{(\mathbf{a}_{k,c} \mathbf{a}_{k,c}^H \theta)}{|\theta^H \mathbf{a}_{k,c}|^2} - \frac{\sum_{i \in \mathcal{K}, i \neq k} (\mathbf{a}_{k,i} \mathbf{a}_{k,i}^H \theta) + \sqrt{\Xi_{k,cI}} (\mathbf{a}_{k,c} \mathbf{a}_{k,c}^H \theta)}{\sum_{i \in \mathcal{K}, i \neq k} |\theta^H \mathbf{a}_{k,i}|^2 + \Xi_{k,cI} |\theta^H \mathbf{a}_{k,c}|^2 + \sigma^2}$. Next, $\Delta \hat{R}_{I,\text{sum}}^{FBL}$ is projected orthogonally onto the complex circle to obtain the Riemannian gradient of $\hat{R}_{I,\text{sum}}$ as

$$\begin{aligned} \text{grad} \hat{R}_{I,\text{sum}}^{FBL}(l+1) &= \Delta \hat{R}_{I,\text{sum}}^{FBL}(l+1) \\ &\quad - \Re(\Delta \hat{R}_{I,\text{sum}}^{FBL}(l+1) \circ \theta^*(l+1)) \circ \theta(l+1). \end{aligned} \quad (32)$$

Search Direction : The search direction is defined as the tangent vector conjugate to Riemannian Gradient, which can be evaluated as

$$\mathbf{d}(l+1) = -\text{grad} \hat{R}_{I,\text{sum}}^{FBL}(l+1) + \omega_1(l+1) \mathcal{T}(l+1), \quad (33)$$

where ω_1 denotes the Polak-Ribiere parameter [43, p. 42] and $\mathcal{T}$ represents the transport function obtained using

$$\mathcal{T}(l+1) = \mathbf{d}(l) - \Re(\mathbf{d}(l) \circ \theta^*(l+1)) \circ \theta(l+1). \quad (34)$$

Updating θ (Retraction): To update the optimization variable, the tangent vector, called a *transport*, is mapped from one tangent space to another. In other words, this transport vector is projected back to the complex circle manifold and the new θ is updated as

$$\theta(l+1) \leftarrow \frac{(\theta(l) + \omega_2(l) \mathbf{d}(l))}{|\theta(l) + \omega_2(l) \mathbf{d}(l)|}, \quad (35)$$

where ω_2 denotes the Armijo backtracking line search step size calculated using [44]. Thus, using the above steps, Algorithm 1 presents a solution to the optimization problem (P2) using the RCG method and provides a sub-optimal θ for fixed coordinates of UAV/RIS. The total computation cost in Algorithm 1 is majorly dominated by the calculation of retraction and gradient in step 7 and step 9 with a complexity of $\mathcal{O}(K^2N)$ and $\mathcal{O}(K^2N^2)$, respectively. Therefore, Algorithm 1 solves the optimization problem (P2) with a computational complexity of $\mathcal{O}(K^2N) + \mathcal{O}(K^2N^2)$. However, considering the fact that $\mathcal{O}(K^2N) \ll \mathcal{O}(K^2N^2)$ for large N , the computational complexity of Algorithm 1 can be approximated as $\mathcal{O}(K^2N^2)$.

Algorithm 1 RCG Algorithm for Optimal Phase Shift Matrix

- 1: Initialize $a_{i,k} \forall k \in \mathcal{K}$
 - 2: Initialize $b_{i,k} \forall k \in \mathcal{K}$
 - 3: Initialize $l = 0$
 - 4: Evaluate $\mathbf{d}(0) = -\text{grad} \hat{R}_{I,\text{sum}}^{FBL}(0)$ using (32)
 - 5: **repeat**
 - 6: Evaluate $\omega_2(l)$
 - 7: Update $\theta(l+1)$ using (35)
 - 8: Evaluate $\mathbf{A}_{k_c}(l+1)$ and $\mathbf{A}_{k_p}(l+1)$
 - 9: Evaluate $\text{grad} \hat{R}_{I,\text{sum}}^{FBL}(l+1)$ using (32)
 - 10: Choose $\omega_1(l+1)$
 - 11: Evaluate $\mathcal{T}(l+1)$ using (34)
 - 12: Evaluate $\mathbf{d}(l+1)$ using (33)
 - 13: $l \leftarrow (l+1)$
 - 14: **until** Convergence is achieved
-

Algorithm 2 Sub-Optimal position of UAV-BS

- 1: Initialize $x = x_{\min} : 1 : x_{\max}$, $z = z_{\min} : 1 : z_{\max}$, $y = y_{\min} : 1 : y_{\max}$,
 - 2: **repeat**
 - 3: **for** $p = 1 : \text{length}(z)$ **do** $z_u = z(p)$
 Calculate $\text{temp}_1(p) = \hat{R}_{I,\text{sum}}^{FBL}(z_u, y_u, x_u)$
 - 4: **end for**
 - 5: Evaluate $\hat{z}_u = z(\arg \max_p(\text{temp}_1(p)))$
 - 6: **for** $q = 1 : \text{length}(y)$ **do** $y_u = y(q)$
 Calculate $\text{temp}_2(q) = \hat{R}_{I,\text{sum}}^{FBL}(\hat{z}_u, y_u, x_u)$
 - 7: **end for**
 - 8: Evaluate $\hat{y}_u = y(\arg \max_q(\text{temp}_2(q)))$
 - 9: **for** $r = 1 : \text{length}(x)$ **do** $x_u = x(r)$
 Calculate $\text{temp}_3(r) = \hat{R}_{I,\text{sum}}^{FBL}(\hat{z}_u, \hat{y}_u, x_u)$
 - 10: **end for**
 - 11: Evaluate $\hat{x}_u = x(\arg \max_r(\text{temp}_3(r)))$
 - 12: $z_u = \hat{z}_u$, $y_u = \hat{y}_u$, $x_u = \hat{x}_u$
 - 13: **until** Convergence is achieved
-

Sub-Optimal UAV/RIS position for fixed Phase shift matrix: For fixed phase shift matrix at RIS, the optimization problem (P0) can be reformulated in a sub-optimal problem for finding the position of UAV/RIS to maximize the weighted sum-rate as follows:

$$(P3) \quad \max_{x_u, y_u, z_u} \hat{R}_{I,\text{sum}}^{FBL}, \quad (36)$$

$$s.t. (26) \text{ of } (P0). \quad (37)$$

Similar to [33], the optimal solution to problem (P3) can be obtained using three dimensional search based brute-force (BF) algorithm. However, due to use of parallel three dimensional search, the BF algorithm has a very high computational complexity. Therefore, we propose Algorithm 2 that finds sub-optimal value of each coordinate $\{x_u, y_u, z_u\}$ separately using one dimensional search. Unlike BF, it uses sequential approach and find a sub-optimal value of each coordinated for a fixed value of other two coordinates. Thus, it solves the problem (P3) with an approximate complexity of $\mathcal{O}(C_1)$, where C_1 is the maximum of length of x , y and z . Now, the complexity of BF-algorithm is approximately $\mathcal{O}(C_1^3)$ [27]. Therefore, considering $\mathcal{O}(C_1^3) \gg \mathcal{O}(C_1)$, Algorithm 2 provides solution similar to BF-algorithm but with a much lesser complexity.

Finally, we determine the $R_{ic}^{FBL} \forall i \in \mathcal{K}$ using a similar method that was used to obtain sub-optimal power coefficients. For the same, we simplify the constraint on the common rate distribution at each user as $\sum_{i \in \mathcal{K}} a_i = 1$, where $a_i \in (0, 1)$ and $R_{ic}^{FBL} = (a_i/\eta_i)R_{k,cI}^{FBL}$. Then we form a sample space $\mathcal{D}$ which is a matrix of size $D \times K$ with D representing the total possible combination of values of $\{a_1, a_2, \dots, a_K\}$ that satisfy (24). Note that i^{th} column of $\mathcal{D}$ represents all the possible values of a_i and sum of each row is equal to unity.

Joint Optimization of UAV/RIS-Position and Phase Shift Matrix: Finally, we proposed Algorithm 3 that maximizes the weighted sum rate by using the aforementioned BCD method⁹ and jointly determines the optimum power allocation coefficients at BS, phase shift matrix at RIS, UAV/RIS-position and common rate splitting corresponding to each user. It solves the main problem with a computational complexity of $i_{max}\mathcal{O}(V + K^2N^2 + C_1) + \mathcal{O}(D)$, where i_{max} is the maximum number of iterations required to solve the problem and $\mathcal{O}(D)$ is the computations required to obtain optimal $\{a_c, a_1, a_2, \dots, a_K\}$.

2) **UAV-aided Transmission:** Using (12) and (15), the achievable rate at k^{th} user while decoding common message for FBL messaging using UAV-aided mode can be expressed as

$$R_{k,cu}^{FBL} \approx \log_2(1 + S_{k,cu}^{\min}) - Q^{-1}(\tilde{\xi}_{kc})(\log_2 e / \sqrt{r_c}) \times \sqrt{1 - (1 + S_{k,cu}^{\min})^{-2}}, \quad (38)$$

where $S_{k,cu}^{\min} = \min(S_{cu}, \hat{S}_{k,cu})$. Similarly, the achievable rate at k^{th} user while decoding a private message can be expressed as

$$R_{k,pu}^{FBL} \approx \log_2(1 + S_{ku}^{\min}) - Q^{-1}(\tilde{\xi}_{kp})(\log_2 e / \sqrt{r_{kp}}) \times \sqrt{1 - (1 + S_{ku}^{\min})^{-2}}, \quad (39)$$

where $S_{ku}^{\min} = \min(S_{ku}, \hat{S}_{ku})$. Using (38) and (39), the total weighted sum rate of the considered system using UAV-aided

⁹Note that, apart from θ , the optimum values of $x_u, y_u, z_u, \alpha_c, \alpha_k \forall k \in \mathcal{K}$ can be obtained for any value of SINR using the proposed method. However, in order to obtain optimum θ , other optimization methods such as Nelder-Mead simplex (NMS) [45], deep neural network (DNN) framework [46], etc., can be applied.

Algorithm 3 BCD-based Joint Optimization

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1: Initialize  $\tilde{\xi}_{kc}^{\max}, \tilde{\xi}_{kp}^{\max}, \eta_k \forall k \in \mathcal{K}, P_b, \mathcal{P}, \mathcal{D}, i = 0, \tau, z_u, y_u,$  and  $x_u$ 
2: repeat
3:    $i \rightarrow i + 1$ 
4:   for  $j = 1 : V$  do  $\{\alpha_c, \alpha_1, \alpha_2, \dots, \alpha_K\} = \mathcal{P}(j, :)$ 
5:     Calculate  $\text{temp}_0(j) = \hat{R}_{I,\text{sum}}^{FBL}$  using (29)
6:   end for
7:   Update  $\{\alpha_c, \alpha_1, \alpha_2, \dots, \alpha_K\} = \mathcal{P}(\arg \max_j(\text{temp}_0(j)), :)$ 
8:   Obtain  $\theta$  using Algorithm 1
9:   Obtain  $\{x_u, y_u, z_u\}$  using Algorithm 2
10:  Evaluate  $\hat{R}_{I,\text{sum}}^{FBL}(i)$  using (29)
11: until  $i > 2$  and  $\hat{R}_{I,\text{sum}}^{FBL}(i) - \hat{R}_{I,\text{sum}}^{FBL}(i-1) < \tau$ 
12: Obtain final  $\{x_u, y_u, z_u\}, \theta$  and  $\{\alpha_c, \alpha_1, \alpha_2, \dots, \alpha_K\}$ 
13: for  $m = 1 : D$  do  $\{a_c, a_1, a_2, \dots, a_K\} = \mathcal{D}(m, :)$ 
14:   Calculate  $\text{temp}_1(m) = R_{I,\text{sum}}^{FBL}$  using (19)
15: end for
16: Update  $\{a_c, a_1, a_2, \dots, a_K\} = \mathcal{D}(\arg \max_m(\text{temp}_1(m)), :)$ 
17: Obtain  $R_{ic}^{FBL} = (a_i/\eta_i)R_{k,cI}^{FBL} \forall i \in \mathcal{K}$ 

```

transmission can be expressed as

$$R_{u,\text{sum}}^{FBL} = \sum_{k \in \mathcal{K}} \eta_k (\hat{R}_{kc}^{FBL} + R_{k,pu}^{FBL}) = \min_{k \in \mathcal{K}} R_{k,cu}^{FBL} + \sum_{k \in \mathcal{K}} \eta_k R_{k,pu}^{FBL}, \quad (40)$$

where $\hat{R}_{kc}^{FBL}$ is the rate allocated to GU- k . Similar to RIS-aided transmission, the weighted sum-rate maximization problem for UAV-aided transmission mode can be formulated as

$$(P4) \quad \max_{\substack{\tilde{\xi}_{kc}, \tilde{\xi}_{kp}, x_u, y_u, z_u, \\ \alpha_c, \alpha_k, \hat{R}_{kc}^{FBL} \forall k \in \mathcal{K}}} R_{u,\text{sum}}^{FBL} \quad (41)$$

$$\text{s.t. } \tilde{\xi}_{kc} \leq \tilde{\xi}_{kc}^{\max}, \tilde{\xi}_{kp} \leq \tilde{\xi}_{kp}^{\max} \forall k \in \mathcal{K} \quad (42)$$

$$\alpha_c + \sum_{k \in \mathcal{K}} \alpha_k = 1 \quad (43)$$

$$\sum_{k \in \mathcal{K}} \eta_k \hat{R}_{kc}^{FBL} \leq \min_{k \in \mathcal{K}} R_{k,cu}^{FBL} \quad (44)$$

$$\hat{R}_{kc}^{FBL} + R_{k,pu}^{FBL} \geq R_{k,\min} \forall k \in \mathcal{K} \quad (45)$$

$$x_{\min} \leq x_u \leq x_{\max}, y_{\min} \leq y_u \leq y_{\max}, \\ z_{\min} \leq z_u \leq z_{\max}. \quad (46)$$

Note that $R_{u,\text{sum}}^{FBL}$ is independent of θ , thus for any range of SINR, the sum rate maximization in problem (P4) is similar to the problem (P3). Therefore, it can be solved using Algorithm 3 for $R_{u,\text{sum}}^{FBL}$ instead of $R_{I,\text{sum}}^{FBL}$ to find optimal coordinates that maximizes the sum rate using UAV-aided mode.

3) **Hybrid RIS-FD-UAV Transmission Mode:** In this mode, each user applies SINR-based SC to select the transmission mode. Thus, the achievable sum rate in this mode can be evaluated as the rate of the selected transmission mode which is given by

$$R_{h,\text{sum}}^{FBL} = \max(R_{u,\text{sum}}^{FBL}, R_{I,\text{sum}}^{FBL}), \quad (47)$$

where $R_{I,\text{sum}}^{FBL}$ and $R_{u,\text{sum}}^{FBL}$ are the given by (19) and (40), respectively. Further, depending on the selected transmission mode, the weighted sum rate can be maximized by solving either problem (P0) or problem (P4) as discussed in previous sub-sections.

Remark 1. The weighted sum rate considering IBL messaging can be obtained by substituting blocklength as infinity in (19), (40), and (47) for RIS-aided, UAV-aided, and hybrid RIS-FD-UAV-aided transmission modes, respectively. Therefore, the formulation and solution of the weighted sum rate maximization problem for IBL messaging will be similar to the FBL case, the details are omitted due to the paucity of space.

IV. FINITE BLOCKLENGTH ANALYSIS

In this section, we analyze the reliability of the considered network in terms of BLER and achievable rate at each user using the optimal position and phase shift matrix obtained in the previous section considering FBL messaging for each mode of transmission.

1) *RIS-aided Transmission:* The average BLER at k^{th} GU for decoding common message for sufficiently large block-length can be approximated as [38]

$$\xi_{k,cI} \approx \mathbb{E} \left[Q \left(\frac{\log_2(1 + S_{k,cI}) - \rho_c}{\sqrt{\Lambda_{k,c}/r_c}} \right) \right], \quad (48)$$

where $\rho_c = \kappa_c/r_c$ with common message having κ_c number of information bits, and $\Lambda_{k,cu} = (\log_2 e)^2 (1 - (1 + S_{k,cI})^{-2})$. After decoding the common message, each user applies SIC and decodes its private message with an average BLER of

$$\xi_{k,pI} \approx \mathbb{E} \left[Q \left(\frac{\log_2(1 + S_{k,I}) - \rho_{kp}}{\sqrt{\Lambda_{k,I}/r_{kp}}} \right) \right], \quad (49)$$

where $\rho_{kp} = \kappa_{kp}/r_{kp}$ with private message having κ_{kp} number of information bits, and $\Lambda_{k,I} = (\log_2 e)^2 (1 - (1 + S_{k,I})^{-2})$. From (48) and (49), the average BLER at each user (ξ_i) will be the mean of individual BLERs of both common and private messages given by

$$\xi_{kI} = (\xi_{k,cI} + \xi_{k,pI})/2. \quad (50)$$

The closed-form expression of the average BLER at each user is derived in the following theorem.

Theorem 1. The closed-form expression of the average BLER at each user (ξ_{kI}) for RIS-aided mode with imperfect SIC is obtained in (51) on the top of the next page, where $\vartheta_c = \delta_c - \frac{1}{2\lambda_c\sqrt{r_c}}$, $\nu_c = \delta_c + \frac{1}{2\lambda_c\sqrt{r_c}}$ with $\delta_c = 2^{\rho_c} - 1$, $\lambda_{kp} = \frac{1}{2\pi\sqrt{2^{2\rho_{kp}}-1}}$, $\vartheta_{kp} = \delta_{kp} - \frac{1}{2\lambda_{kp}\sqrt{r_{kp}}}$, and $\nu_{kp} = \delta_{kp} + \frac{1}{2\lambda_{kp}\sqrt{r_{kp}}}$ with $\delta_{kp} = 2^{\rho_{kp}} - 1$.

Proof. Solving and finding a closed-form of (48) is analytically intractable. Thus, similar to [47], we use approximation $\xi_{k,cI} \approx \mathbb{E} [\Upsilon_{k,cI}(S_{k,cI})]$ with

$$\Upsilon_{k,cI}(S_{k,cI}) = \begin{cases} 1, & S_{k,cI} \leq \vartheta_c, \\ 0.5 - \lambda_c\sqrt{r_c}(S_{k,cI} - \delta_c), & \vartheta_c \leq S_{k,cI} \leq \nu_c, \\ 0, & S_{k,cI} \geq \nu_c. \end{cases} \quad (52)$$

Using probability distribution function (PDF) of $S_{k,cI}$, we can write

$$\xi_{k,cI} \approx \int_0^\infty \Upsilon_{k,cI}(x) f_{S_{k,cI}}(x) dx. \quad (53)$$

Solving (53) using integration by parts, we have

$$\xi_{k,cI} \approx \lambda_c\sqrt{r_c} \int_{\vartheta_c}^{\nu_c} F_{S_{k,cI}}(x) dx. \quad (54)$$

Next, the involved integration in the above equation can be solved using Riemann integral approximation $\int_a^b g(y) dy = (b-a)g((a+b)/2)$. Thus, BLER in (54) can be expressed as

$$\xi_{k,cI} \approx \lambda_c\sqrt{r_c}(\nu_c - \vartheta_c) F_{S_{k,cI}} \left(\frac{\nu_c + \vartheta_c}{2} \right). \quad (55)$$

To solve (55), we find the cumulative distribution function (CDF) of the SINR $S_{k,cI}$ in terms of CDF of channel gain $|g_{kIb}|^2$ as

$$F_{S_{k,cI}}(x) = F_{|g_{kIb}|^2} \left(\frac{x\sigma^2}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i x) P_b} \right). \quad (56)$$

Using (55) and (56), the BLER can be expressed as

$$\xi_{k,cI} \approx \lambda_c\sqrt{r_c}(\nu_c - \vartheta_c) \times F_{|g_{kIb}|^2} \left(\frac{(\nu_c + \vartheta_c)\sigma^2}{(2\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i(\nu_c + \vartheta_c)) P_b} \right). \quad (57)$$

To find the CDF $F_{|g_{kIb}|^2}(x)$, we first find the PDF of $Z_n = |g_{kn}g_{nb}|$ as follows. Since the channels g_{kn} and g_{nb} are Rician distributed, therefore the PDF of Z_n can be evaluated as [48]

$$f_{Z_n}(z) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \frac{4z^{i+j+1}(\mathbf{b}+1)^{i+j+2}\mathbf{b}^{i+j}}{(i!j!e^{\mathbf{b}})^2(\beta_{kI}\beta_{Ib})^{-\frac{i+j+2}{2}}} \times K_{i-j} \left(\frac{2z(\mathbf{b}+1)}{\sqrt{\beta_{kI}\beta_{Ib}}} \right). \quad (58)$$

Here $\beta_{nb} \approx \beta_{Ib}$ and $\beta_{kn} \approx \beta_{kI} \forall n \in \mathcal{I}$. Using (58), the mean and variance of Z_n are given by $\mu = \mathbb{E}(Z_n) = \frac{\pi\sqrt{\beta_{kI}\beta_{Ib}}}{4(\mathbf{b}+1)}(L_{\frac{1}{2}}(-\mathbf{b}))^2$, and $\Omega = \beta_{kI}\beta_{Ib}[1 - \frac{\pi^2}{16(\mathbf{b}+1)^2}(L_{\frac{1}{2}}(-\mathbf{b}))^4]$, respectively. Due to analytical complexity of $f_{Z_n}(z)$, obtaining the PDF of $|g_{kIb}| = \sum_{n \in \mathcal{I}} Z_n$ from (58) is extremely difficult and challenging. Therefore, similar to [49], we use μ , Ω , and the properties of the series of Laguerre polynomials to obtain an approximate PDF of $|g_{kIb}|$ given by

$$f_{|g_{kIb}|}(x) = \frac{x^\varpi \exp(-\mu x/\Omega)}{(\Omega/\mu)^{\varpi+1} \Gamma(\varpi+1)}, \quad (59)$$

where $\varpi = (\mu^2 N/\Omega) - 1$. Further, using (59), we evaluate the CDF of $|g_{kIb}|$ as

$$F_{|g_{kIb}|}(x) = \int_0^x \frac{t^\varpi \exp(-\mu t/\Omega)}{(\Omega/\mu)^{\varpi+1} \Gamma(\varpi+1)} dt. \quad (60)$$

Solving (60) using [50], we obtain

$$F_{|g_{kIb}|}(x) = \frac{1}{\Gamma(\varpi+1)} \gamma(\varpi+1, \mu x/\Omega). \quad (61)$$

$$\begin{aligned} \xi_{kI} &\approx \frac{\lambda_c \sqrt{r_c} (\nu_c - \vartheta_c)}{2\Gamma \left[\frac{\pi^2 \chi^4 N}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4} \right]} \gamma \left(\frac{\pi^2 \chi^4 N}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4}, \frac{4\pi \chi^2 (\mathfrak{b}+1)}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4} \right) \\ &\times \sqrt{\frac{(\nu_c + \vartheta_c) \sigma^2}{(2\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i (\nu_c + \vartheta_c)) \beta_{kI} \beta_{Ib} P_b}} + \frac{\lambda_{kp} \sqrt{r_{kp}} (\nu_{kp} - \vartheta_{kp})}{2\Gamma \left[\frac{\pi^2 \chi^4 N}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4} \right]} \gamma \left(\frac{\pi^2 \chi^4 N}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4}, \right. \\ &\left. \frac{4\pi \chi^2 (\mathfrak{b}+1)}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4} \sqrt{\frac{(\nu_{kp} + \vartheta_{kp}) \sigma^2}{(2\alpha_k - (\sum_{j \in \mathcal{K}, j \neq k} \alpha_j + \Xi_{k,cI} \alpha_c) (\nu_{kp} + \vartheta_{kp})) \beta_{kI} \beta_{Ib} P_b}} \right). \end{aligned} \quad (51)$$

Further, using (61), the CDF of $|g_{kIb}|^2$ can be expressed as

$$\begin{aligned} F_{|g_{kIb}|^2}(x) &= F_{|g_{kIb}|}(\sqrt{x}) \\ &= \frac{1}{\Gamma(\varpi + 1)} \gamma(\varpi + 1, \mu \sqrt{x}/\Omega). \end{aligned} \quad (62)$$

Furthermore, similar to (53), we can write

$$\xi_{k,pI} \approx \int_0^\infty \Theta_{kI}(x) f_{S_{kI}}(x) dx, \quad (63)$$

where $f_{S_{kI}}(\cdot)$ is the PDF of S_{kI} and $\Theta_{kI}(\cdot)$ can be expressed as

$$\Theta_{kI}(S_{kI}) = \begin{cases} 1, & \hat{S}_{kI} \leq \vartheta_{kp}, \\ 0.5 - \lambda_{kp} \sqrt{r_{kp}} (S_{kI} - \delta_{kp}), & \vartheta_{kp} \leq S_{kp} \leq \nu_{kp}, \\ 0, & S_{kp} \geq \nu_{kp}. \end{cases} \quad (64)$$

Similar to (55) and (57), BLER in (63) can be expressed as

$$\begin{aligned} \xi_{k,pI} &\approx \lambda_{kp} \sqrt{r_{kp}} (\nu_{kp} - \vartheta_{kp}) \\ &\times F_{|g_{kIb}|^2} \left(\frac{(\nu_{kp} + \vartheta_{kp}) \sigma^2 / P_b}{2\alpha_k - (\sum_{j \in \mathcal{K}, j \neq k} \alpha_j + \Xi_{k,cI} \alpha_c) (\nu_{kp} + \vartheta_{kp})} \right). \end{aligned} \quad (65)$$

Obtaining $\xi_{k,cI}$ and $\xi_{k,pI}$ using (57), (62), and (65), and substituting in (50), we obtain closed-form expression of average BLER at each user in (51). ■

Further, the average weighted sum rate after decoding common and private messages at each user considering imperfect SIC with FBL is obtained using (19) as $\tilde{R}_{I,\text{sum}}^{FBL} = \mathbb{E}[R_{I,\text{sum}}^{FBL}] = \min_{k \in \mathcal{K}} \eta_k \mathbb{E}[R_{k,cI}^{FBL}] + \sum_{k \in \mathcal{K}} \eta_k \mathbb{E}[R_{k,pI}^{FBL}]$. The closed-form expression of $\tilde{R}_{I,\text{sum}}^{FBL}$ is evaluated in the following theorem.

Theorem 2. *The average weighted sum rate at each user after decoding common and private messages with FBL messaging considering imperfect SIC is obtained in (66) on the top of the next page, where $\mathbb{F}_p(x) = \Gamma \left(\mathfrak{c}, \frac{4\pi \chi^2 (\mathfrak{b}+1)}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4} \sqrt{\frac{x \sigma^2 / ((\beta_{kI} \beta_{Ib} P_b))}{(\alpha_k - (\sum_{j \in \mathcal{K}, j \neq k} \alpha_j + \Xi_{k,cI} \alpha_c) x)} \right), \mathbb{F}_c(x) = \Gamma \left(\mathfrak{c}, \frac{4\pi \chi^2 (\mathfrak{b}+1)}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4} \sqrt{\frac{x \sigma^2 / ((\beta_{kI} \beta_{Ib} P_b))}{(\alpha_c - \sum_{j \in \mathcal{K}} \alpha_j x)} \right), \mathfrak{c} = \frac{\pi^2 \chi^4 N}{16(\mathfrak{b}+1)^2 - \pi^2 \chi^4}, \mathcal{M} = \{M_1, M_2, M_3, M_4\}$ denote the respective accuracy-complexity trade-off factors, t_n represents n^{th} root of the generalized Laguerre polynomial $L_{m \in \mathcal{M}}(\cdot)$, $w_n = \frac{t_n}{(m+1)^2 [L_{m+1}^\varphi(t_n)]^2}$ represents the weight vector.*

Proof. Solving $\mathbb{E}[R_{k,cI}^{FBL}]$ w.r.t. PDF of $S_{k,cI}$, we have

$$\begin{aligned} \mathbb{E}[R_{k,cI}^{FBL}] &= \int_0^\infty \left(\log_2(1+x) - C_c \sqrt{(1 - ((1+x)^{-2}))} \right) \\ &\times f_{S_{k,cI}}(x) dx \\ &= \int_0^\infty \left(\frac{1}{1+x} - \frac{C_c}{(1+x)^2 (\sqrt{x^2 + 2x})} \right) \\ &\times (1 - F_{S_{k,cI}}(x)) dx. \end{aligned} \quad (67)$$

where $C_c = Q^{-1}(\tilde{\xi}_{kc}) \log_2 e / \sqrt{r_c}$. Similarly, $\mathbb{E}[R_{k,pI}^{FBL}]$ can also be simplified as

$$\begin{aligned} \mathbb{E}[R_{k,pI}^{FBL}] &= \int_0^\infty \left(\frac{1}{1+x} - \frac{Q^{-1}(\tilde{\xi}_{kp}) \log_2 e / \sqrt{r_{kp}}}{(1+x)^2 (\sqrt{x^2 + 2x})} \right) \\ &\times (1 - F_{S_{k,pI}}(x)) dx. \end{aligned} \quad (68)$$

Note that substituting $F_{S_{k,cI}}$ from (56) into (67) and $F_{S_{k,pI}}$ from (65) into (68) results in analytically intractable expressions. Thus, solving these equations using standard mathematical methods becomes extremely difficult. Therefore, we use Gaussian-Laguerre quadrature method [51] to solve (67) and (68), and obtain (66). ■

2) *UAV-aided Transmission:* Similar to RIS-aided mode, the average BLER at k^{th} GU for decoding common messages using UAV-aided transmission can be approximated as

$$\xi_{k,cu} \approx \mathbb{E} \left[Q \left(\frac{\log_2(1 + S_{k,cu}^{\min}) - \rho_c}{\sqrt{\Lambda_{k,cu} / r_c}} \right) \right], \quad (69)$$

where $\Lambda_{k,cu} = (\log_2 e)^2 (1 - (1 + S_{k,cu}^{\min})^{-2})$. After decoding the common message, each user applies SIC and decodes its private message with an average BLER of

$$\xi_{k,pu} \approx \mathbb{E} \left[Q \left(\frac{\log_2(1 + S_{ku}^{\min}) - \rho_{ku}}{\sqrt{\Lambda_{ku} / r_{ku}}} \right) \right], \quad (70)$$

where $\Lambda_{ku} = (\log_2 e)^2 (1 - (1 + S_{ku}^{\min})^{-2})$. Note that $S_{k,cu}^{\min}$ and $S_{ku}^{\min}$ are defined in (38) and (39), respectively. From (69) and (70), the total average BLER at each user is obtained as

$$\xi_{ku} = (\xi_{k,cu} + \xi_{k,pu}) / 2. \quad (71)$$

The closed-form expression of the average BLER at each user is derived in the following theorem.

$$\begin{aligned} \tilde{R}_{I,\text{sum}}^{FBL} \approx & \min_{k \in \mathcal{K}} \frac{1}{\ln 2 \Gamma \left[\frac{\pi^2 \chi^4 N}{16(b+1)^2 - \pi^2 \chi^4} \right]} \left(\sum_{i=1}^{M_1} \frac{\eta_k w_{c,i} e^{t_{c,i}} \mathbb{F}_c(t_{c,i})}{(1+t_{c,i})} - \sum_{j=1}^{M_2} \frac{\eta_k Q^{-1}(\tilde{\xi}_{kc}) w_{ec,j} e^{t_{ec,j}} \mathbb{F}_c(t_{ec,j})}{\sqrt{r_c}(1+t_{ec,j})^2 \sqrt{t_{ec,j}^2 + 2t_{ec,j}}} \right) \\ & + \sum_{k \in \mathcal{K}} \frac{1}{\ln 2 \Gamma \left[\frac{\pi^2 \chi^4 N}{16(b+1)^2 - \pi^2 \chi^4} \right]} \left(\sum_{l=1}^{M_3} \frac{\eta_k w_{p,l} e^{t_{p,l}} \mathbb{F}_p(t_{p,l})}{(1+t_{p,l})} - \sum_{r=1}^{M_4} \frac{\eta_k Q^{-1}(\tilde{\xi}_{kp}) w_{ep,r} e^{t_{ep,r}} \mathbb{F}_p(t_{ep,r})}{\sqrt{r_{kp}}(1+t_{ep,r})^2 \sqrt{t_{ep,r}^2 + 2t_{ep,r}}} \right). \end{aligned} \quad (66)$$

Theorem 3. The closed-form expression of the total average BLER at each user (ξ_{ku}) for UAV-aided mode with FBL messaging considering imperfect SIC is obtained as

$$\begin{aligned} \xi_{ku} \approx & (1/2) (\lambda_c \sqrt{r_c} (\nu_c - \vartheta_c) [1 - \mathbb{F}_{cu}(\nu_c + \vartheta_c)] \\ & + \lambda_{kp} \sqrt{r_{kp}} (\nu_{kp} - \vartheta_{kp}) [1 - \mathbb{F}_{pu}(\nu_{kp} + \vartheta_{kp})]), \end{aligned} \quad (72)$$

where $\mathbb{F}_{cu}(x)$ and $\mathbb{F}_{pu}(x)$ are given in (73) and (73), respectively, on the top of the next page.

Proof. Solving and finding a closed-form of (69) is analytically intractable. Thus, similar to [47], we can approximate $\xi_{k,cu}$ as

$$\xi_{k,cu} \approx \mathbb{E} [\Upsilon_{k,cu}(S_{k,cu}^{\min})], \quad (75)$$

where $\Upsilon_{k,cu}(S_{k,cu}^{\min})$ is an approximation given by

$$\Upsilon_{k,cu}(S_{k,cu}^{\min}) = \begin{cases} 1, & S_{k,cu}^{\min} \leq \vartheta_c, \\ 0.5 - \lambda_c \sqrt{r_c} (S_{k,cu}^{\min} - \delta_c), & \vartheta_c \leq S_{k,cu}^{\min} \leq \nu_c, \\ 0, & S_{k,cu}^{\min} \geq \nu_c. \end{cases} \quad (76)$$

Further simplifying (75), we obtain

$$\xi_{k,cu} \approx \int_0^\infty \Upsilon_{k,cu}(x) f_{S_{k,cu}^{\min}}(x) dx. \quad (77)$$

Solving (77) using integration by parts, we have

$$\xi_{k,cu} \approx \lambda_c \sqrt{r_c} \int_{\vartheta_c}^{\nu_c} F_{S_{k,cu}^{\min}}(x) dx. \quad (78)$$

Thus, BLER in (78) can be expressed as

$$\xi_{k,cu} \approx \lambda_c \sqrt{r_c} (\nu_c - \vartheta_c) F_{S_{k,cu}^{\min}} \left(\frac{\nu_c + \vartheta_c}{2} \right). \quad (79)$$

Similar to (55), (79) can be simplified as (80) on the top of the next page. Using CDF of Rician distribution [52] and solving (80), we obtain

$$\xi_{k,cu} \approx \lambda_c \sqrt{r_c} (\nu_c - \vartheta_c) [1 - \mathbb{F}_{cu}(\nu_c + \vartheta_c)]. \quad (81)$$

Further, similar to (77), we can write

$$\xi_{k,pu} \approx \int_0^\infty \Theta_{ku}(x) f_{S_{ku}^{\min}}(x) dx, \quad (82)$$

with

$$\Theta_{ku}(\hat{S}_{ku}) = \begin{cases} 1, & S_{ku}^{\min} \leq \vartheta_{kp}, \\ 0.5 - \lambda_{kp} \sqrt{r_{kp}} (S_{ku}^{\min} - \delta_{kp}), & \vartheta_{kp} \leq S_{ku}^{\min} \leq \nu_{kp}, \\ 0, & S_{ku}^{\min} \geq \nu_{kp}. \end{cases} \quad (83)$$

Similar to (79), BLER in (82) can be expressed as

$$\xi_{k,cu} \approx \lambda_{kp} \sqrt{r_{kp}} (\nu_{kp} - \vartheta_{kp}) [1 - \mathbb{F}_{pu}(\nu_{kp} + \vartheta_{kp})]. \quad (84)$$

Substituting (81) and (84) in (71), we obtain (72). ■

Further, the average weighted sum rate for UAV-aided mode at each user considering imperfect SIC is obtained using (40) as $\tilde{R}_{u,\text{sum}}^{FBL} = \mathbb{E}[R_{u,\text{sum}}^{FBL}] = \min_{k \in \mathcal{K}} \eta_k \mathbb{E}[R_{k,cu}^{FBL}] + \sum_{k \in \mathcal{K}} \eta_k \mathbb{E}[R_{k,pu}^{FBL}]$. The closed-form expression of $\tilde{R}_{u,\text{sum}}^{FBL}$ is evaluated in the following theorem.

Theorem 4. The average weighted sum rate considering imperfect SIC with FBL messaging using UAV-aided mode is obtained in (85) on the top of the next page, where $t_{(\cdot)}$, $w_{(\cdot)}$ and $M_{(\cdot)}$ represents the respective parameters given in (93).

Proof. The weighted sum rate in (85) can be obtained by solving $\mathbb{E}[R_{k,cu}^{FBL}]$ and $\mathbb{E}[R_{k,pu}^{FBL}]$ following steps similar to (67) and (68). ■

3) *Hybrid RIS-FD-UAV-aided Transmission Mode:* In this mode, each user applies SINR-based SC to select the transmission mode and decode the desired symbol. Therefore, BLER at each user can be obtained as the minimum of the BLER UAV-aided and RIS-aided modes and is expressed as

$$\xi_k = \min(\xi_{kI}, \xi_{ku}), \forall k \in \mathcal{K}, \quad (86)$$

where ξ_{kI} and ξ_{ku} are derived in (51) and (72), respectively. Next, the achievable weighted sum rate using this mode is given by

$$\tilde{R}_{\text{sum}}^{FBL} = \max(\tilde{R}_{I,\text{sum}}^{FBL}, \tilde{R}_{u,\text{sum}}^{FBL}), \quad (87)$$

where $\tilde{R}_{u,\text{sum}}^{FBL}$ and $\tilde{R}_{I,\text{sum}}^{FBL}$ are the rates obtained for UAV-aided and RIS-aided mode, respectively.

V. INFINITE BLOCKLENGTH ANALYSIS

This section analyzes the performance of the considered network in terms of outage probability and ergodic capacity using the optimal position and phase shift matrix obtained in Section III with IBL messaging for different transmission modes.

1) *RIS-aided Transmission:* In this mode, RIS reflects the received information from BS to all the users. Let us consider $s_{k,cI}^{th} = 2^{R_{k,cI}} - 1$ and $s_{kI}^{th} = 2^{R_{kI}} - 1$ be the minimum required SNR threshold corresponding to the target rate of $R_{k,cI}$ and R_{kI} corresponding to common and private messages, respectively, at GU- k for decoding its desired symbol.

$$\mathbb{F}_{cu}(x) = \max \left(Q \left(\sqrt{2b}, \sqrt{\frac{2(b+1)x\sigma^2}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i x) \beta_{ub} P_b}} \right), Q \left(\sqrt{2b}, \sqrt{\frac{2(b+1)x\sigma^2}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i x) \beta_{ku} P_u}} \right) \right). \quad (73)$$

$$\mathbb{F}_{pu}(x) = \max \left(Q \left(\sqrt{2b}, \sqrt{\frac{2(b+1)x\sigma^2/(\beta_{ku} P_u)}{(\alpha_k - (\sum_{i \in \mathcal{K}, i \neq k} \alpha_i + \Xi_{k,cu} \alpha_c) x)}} \right), Q \left(\sqrt{2b}, \sqrt{\frac{2(b+1)x\sigma^2/(\beta_{ub} P_b)}{(\alpha_k - (\sum_{i \in \mathcal{K}, i \neq k} \alpha_i + \Xi_{cu} \alpha_c) x)}} \right) \right). \quad (74)$$

$$\xi_{k,cu} \approx \begin{cases} \lambda_c \sqrt{r_c} (\nu_c - \vartheta_c) F_{|g_{ub}|^2} \left(\frac{(\nu_c + \vartheta_c) \sigma^2}{(2\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i (\nu_c + \vartheta_c)) P_u} \right), & S_{cu} \leq \hat{S}_{k,cu} \\ \lambda_c \sqrt{r_c} (\nu_c - \vartheta_c) F_{|g_{ku}|^2} \left(\frac{(\nu_c + \vartheta_c) \sigma^2}{(2\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i (\nu_c + \vartheta_c)) P_u} \right), & S_{cu} > \hat{S}_{k,cu}. \end{cases} \quad (80)$$

$$\begin{aligned} \tilde{R}_{u,\text{sum}}^{FBL} \approx & \min_{k \in \mathcal{K}} \frac{1}{\ln 2} \left(\sum_{i=1}^{M_5} \frac{\eta_k w_{c,i} e^{t_{c,i}} \mathbb{F}_c(t_{c,i})}{(1 + t_{c,i})} - \sum_{j=1}^{M_6} \frac{\eta_k Q^{-1}(\tilde{\xi}_{kc}) w_{ec,j} e^{t_{ec,j}} \mathbb{F}_c(t_{ec,j})}{\sqrt{r_c} (1 + t_{ec,j})^2 \sqrt{t_{ec,j}^2 + 2t_{ec,j}}} \right) \\ & + \sum_{k \in \mathcal{K}} \frac{1}{\ln 2} \left(\sum_{l=1}^{M_7} \frac{\eta_k w_{p,l} e^{t_{p,l}} \mathbb{F}_p(t_{p,l})}{(1 + t_{p,l})} - \sum_{r=1}^{M_8} \frac{\eta_k Q^{-1}(\tilde{\xi}_{kp}) w_{ep,r} e^{t_{ep,r}} \mathbb{F}_p(t_{ep,r})}{\sqrt{r_{kp}} (1 + t_{ep,r})^2 \sqrt{t_{ep,r}^2 + 2t_{ep,r}}} \right), \end{aligned} \quad (85)$$

Using (8) and (9), the outage probability at k^{th} user can be expressed as

$$P_{OKI} = 1 - \Pr(S_{k,cI} > s_{k,cI}^{th}, S_{kI} > s_{kI}^{th}). \quad (88)$$

Theorem 5. The closed-form expression for the outage probability of GU-1 using RIS-aided transmission can be expressed as (89) which is given on the top of the next page, where $\hat{s}_{koI}^{th} = \max \left(\frac{s_{k,cI}^{th}}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i s_{k,cI}^{th})}, \frac{s_{kI}^{th}}{(\alpha_k - (\sum_{j \in \mathcal{K}, j \neq k} \alpha_j + \Xi_{k,cI} \alpha_c) s_{kI}^{th})} \right)$ and $\chi = L_{\frac{1}{2}}(-b)$.

Proof. For the case of exposition, we assume $g_{kIb} = \sum_{n \in \mathcal{I}} g_{kn} g_{nb} \theta_n \forall k \in \mathcal{K}$. Substituting S_{kI} from (9) into (88), we obtain (90) on the top of next page. After some mathematical simplification, (90) can be written as

$$\begin{aligned} \Pr(S_{k,cI} > s_{k,cI}^{th}, S_{kI} > s_{kI}^{th}) &= \Pr(|g_{kIb}|^2 > \tilde{s}_{kI}^{th}) \\ &= 1 - F_{|g_{kIb}|^2}(\tilde{s}_{kI}^{th}), \end{aligned} \quad (91)$$

where $\tilde{s}_{kI}^{th}$ is given on the top of the next page. Substituting $x = \tilde{s}_{kI}^{th}$ in (62) and using (91), we obtain (89). ■

Further, the average weighted sum rate after decoding common and private messages considering imperfect SIC with IBL can be obtained by substituting blocklengths r_c and r_{kp} as infinite in (66) as

$$\begin{aligned} \tilde{R}_{I,\text{sum}}^{IBL} \approx & \min_{k \in \mathcal{K}} \sum_{i=1}^{M_1} \frac{\eta_k w_{c,i} e^{t_{c,i}} \mathbb{F}_c(t_{c,i})}{\ln 2 \Gamma \left[\frac{\pi^2 \chi^4 N}{16(b+1)^2 - \pi^2 \chi^4} \right] (1 + t_{c,i})} \\ & + \sum_{k \in \mathcal{K}} \sum_{l=1}^{M_3} \frac{\eta_k w_{p,l} e^{t_{p,l}} \mathbb{F}_p(t_{p,l})}{(1 + t_{p,l}) \ln 2 \Gamma \left[\frac{\pi^2 \chi^4 N}{16(b+1)^2 - \pi^2 \chi^4} \right]}. \end{aligned} \quad (93)$$

2) *UAV-aided Transmission:* In this mode, UAV operates as an FD relay to forward the information symbols using DF relaying. Let us consider $s_{iu}^{th} = 2^{R_{iu}} - 1$ as the minimum SINR threshold corresponding to the target rate of $R_{iu} \forall i \in \{c, \mathcal{K}\}$ required at UAV. Thus, if $S_{iu} < s_{iu}^{th}$, then there will be an outage at the UAV. Similarly, $\hat{s}_{k,cu}^{th} = 2^{\hat{R}_{k,cu}} - 1$ and $\hat{s}_{ku}^{th} = 2^{\hat{R}_{ku}} - 1$ is the minimum SNR threshold corresponding to the target rate of $\hat{R}_{k,cu} \leq R_{cu}$ and $\hat{R}_{ku} \leq R_{ku}$, respectively, at GU- k required for decoding its desired symbol. Thus, either $\hat{S}_{k,cu} < \hat{s}_{k,cu}^{th}$ or $\hat{S}_{ku} < \hat{s}_{ku}^{th}$ leads to an outage. Therefore, GU- k will be in outage if either of the conditions $S_{iu} < s_{iu}^{th}$ or $\hat{S}_{k,cu} < \hat{s}_{k,cu}^{th}$ or $\hat{S}_{ku} < \hat{s}_{ku}^{th}$ holds true. Hence, using (12) and (16), the outage probability at GU- k can be expressed as

$$P_{OKU} = 1 - \Pr(S_{cu} > s_{cu}^{th}, S_{iu} > s_{iu}^{th} \forall i \in \mathcal{K}, \hat{S}_{k,cu} > \hat{s}_{k,cu}^{th}, \hat{S}_{ku} > \hat{s}_{ku}^{th}). \quad (94)$$

Theorem 6. The closed-form expression for the outage probability at each user using UAV-aided transmission with IBL messaging considering imperfect SIC can be expressed as

$$\begin{aligned} P_{OKU} = & 1 - Q \left(\sqrt{2b}, \sqrt{2(b+1)s_{ou}^{th}(\sigma^2 + P_{SI})/(\beta_{ub} P_b)} \right) \\ & \times Q \left(\sqrt{2b}, \sqrt{2(b+1)\hat{s}_{kou}^{th}\sigma^2/(\beta_{ku} P_u)} \right), \end{aligned} \quad (95)$$

where s_{ou}^{th} and $\hat{s}_{kou}^{th}$ are given in (96) and (97) on the top of the next page.

Proof. The outage probability at GU- k , given in (94), can be expressed as

$$\begin{aligned} P_{OKU} = & 1 - \Pr(S_{cu} > s_{cu}^{th}, S_{iu} > s_{iu}^{th} \forall i \in \mathcal{K}) \\ & \times \Pr(\hat{S}_{k,cu} > \hat{s}_{k,cu}^{th}, \hat{S}_{ku} > \hat{s}_{ku}^{th}). \end{aligned} \quad (98)$$

$$P_{\text{OkI}} = \frac{1}{\Gamma\left[\frac{\pi^2\chi^4 N}{16(\mathfrak{b}+1)^2 - \pi^2\chi^4}\right]} \gamma\left(\frac{\pi^2\chi^4 N}{16(\mathfrak{b}+1)^2 - \pi^2\chi^4}, \frac{4\pi\chi^2(\mathfrak{b}+1)}{16(\mathfrak{b}+1)^2 - \pi^2\chi^4} \sqrt{\frac{\tilde{s}_{koI}^{th}\sigma^2}{\beta_{kI}\beta_{Ib}P_b}}\right). \quad (89)$$

$$\Pr(S_{k,cI} > s_{k,cI}^{th}, S_{kI} > s_{kI}^{th}) = \Pr\left(\frac{|g_{kIb}|^2 \alpha_c P_b}{|g_{kIb}|^2 P_b (\sum_{i \in \mathcal{K}} \alpha_i) + \sigma^2} > s_{k,cI}^{th}, \frac{|g_{kIb}|^2 \alpha_k P_b}{|g_{kIb}|^2 P_b (\sum_{i \in \mathcal{K}, i \neq k} \alpha_i + \Xi_{k,cI} \alpha_c) + \sigma^2} > s_{kI}^{th}\right). \quad (90)$$

$$\tilde{s}_{kI}^{th} = \max\left(\frac{s_{k,cI}^{th}\sigma^2}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i s_{k,cI}^{th})P_b}, \frac{s_{kI}^{th}\sigma^2}{(\alpha_k - (\sum_{j \in \mathcal{K}, j \neq k} \alpha_j + \Xi_{k,cI} \alpha_c) s_{kI}^{th})P_u}\right). \quad (92)$$

$$s_{ou}^{th} = \max\left(\frac{s_{cu}^{th}}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i s_{cu}^{th})}, \frac{s_{iu}^{th}}{(\alpha_i - (\sum_{p \in \mathcal{K}_i} \alpha_p + \sum_{j \in \mathcal{K}_h} \Xi_{ju} \alpha_j + \Xi_{cu} \alpha_c) s_{iu}^{th})} \forall i \in \mathcal{K}\right). \quad (96)$$

$$\hat{s}_{kou}^{th} = \max\left(\frac{\hat{s}_{k,cu}^{th}}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i \hat{s}_{k,cu}^{th})}, \frac{\hat{s}_{ku}^{th}}{(\alpha_k - (\sum_{j \in \mathcal{K}, j \neq k} \alpha_j + \Xi_{cu} \alpha_c) \hat{s}_{ku}^{th})}\right). \quad (97)$$

Substituting S_{1u} and S_{2u} from (12). After some mathematical manipulations, we obtain

$$\Pr(S_{cu} > s_{cu}^{th}, S_{iu} > s_{iu}^{th} \forall i \in \mathcal{K}) = \Pr(|g_{ub}|^2 > \tilde{s}_u^{th}), \quad (99)$$

where $\tilde{s}_u^{th}$ is given in (100) on the top of the next page. Since $|g_{ub}|^2$ follows the Rician distribution, therefore, the expression in (99) can be expressed as

$$\begin{aligned} \Pr(|g_{ub}|^2 > \tilde{s}_u^{th}) &= 1 - \mathbb{F}_{|g_{ub}|^2}(\tilde{s}_u^{th}) \\ &= Q\left(\sqrt{2\mathfrak{b}}, \sqrt{\frac{2(\mathfrak{b}+1)\tilde{s}_u^{th}}{\beta_{ub}}}\right). \end{aligned} \quad (101)$$

Similarly, using (15), we obtain

$$\begin{aligned} \Pr(\hat{S}_{k,cu} > \hat{s}_{k,cu}^{th}, \hat{S}_{ku} > \hat{s}_{ku}^{th}) &= \Pr(|g_{ku}|^2 > \tilde{s}_{ku}^{th}) \\ &= Q\left(\sqrt{2\mathfrak{b}}, \sqrt{\frac{2(\mathfrak{b}+1)\tilde{s}_{ku}^{th}}{\beta_{ku}}}\right), \end{aligned} \quad (102)$$

where $\tilde{s}_{ku}^{th}$ is given in (103) on the top of the next page. Therefore, from (98), (101) and (102), we obtain the closed-form expression for the outage probability of GU- k in (95). ■

Further, the average weighted sum rate of the considered network considering imperfect SIC with IBL can be obtained by substituting blocklengths r_c and r_{kp} as infinite in (85) as

$$\begin{aligned} \tilde{R}_{u,\text{sum}}^{IBL} &\approx \frac{1}{\ln 2} \left(\min_{k \in \mathcal{K}} \sum_{i=1}^{M_5} \frac{\eta_k w_{c,i} e^{t_{c,i}} \mathbb{F}_c(t_{c,i})}{(1+t_{c,i})} \right. \\ &\quad \left. + \sum_{k \in \mathcal{K}} \sum_{l=1}^{M_7} \frac{\eta_k w_{p,l} e^{t_{p,l}} \mathbb{F}_p(t_{p,l})}{(1+t_{p,l})} \right). \end{aligned} \quad (104)$$

3) *Hybrid RIS-FD-UAV-aided Transmission Mode:* The outage probability at each user can be obtained as the product of the individual probability of UAV-aided and RIS-aided modes and is expressed as

$$P_{\text{Ok}} = P_{\text{OkI}} P_{\text{OkU}}, \forall k \in \mathcal{K} \quad (105)$$

where P_{OkI} and P_{OkU} are given in (89) and (95), respectively. Next, the achievable weighted sum rate of the considered network using this mode with IBL messaging can be obtained as

$$\tilde{R}_{\text{sum}}^{IBL} = \max(\tilde{R}_{I,\text{sum}}^{IBL}, \tilde{R}_{u,\text{sum}}^{IBL}), \quad (106)$$

where $\tilde{R}_{u,\text{sum}}^{IBL}$ and $\tilde{R}_{I,\text{sum}}^{IBL}$ are the rates obtained in (89) and (104), respectively.

Remark 2. The average BLER and weighted sum rate with perfect SIC can be obtained by substituting the error in SIC equal to zero in each equation.

VI. RESULTS AND DISCUSSION

This section presents the Monte Carlo simulation-based results to validate the effectiveness of the proposed algorithms. It also validates the accuracy of the obtained closed-form expressions of BLER, outage probability, and weighted sum rate with FBL and IBL messaging of the considered system under perfect and imperfect SIC scenarios, and thus helps us to draw useful insights about the performance characteristics. Without the loss of generality, we consider $K = 2$, and $\Xi_{k,cI} = \Xi_{cu} = \Xi_{ku} = \Xi_{k,cu} = \Xi$, $\eta_k = 1$, $\forall k \in \mathcal{K}$, $P_b = P_u$, and all complexity-trade-off factors are considered to be 70. Further, we consider two users whose coordinates are given by $\{50, 0, 0\}$ and $\{20, 0, 0\}$ whereas coordinates of BS are

$$\tilde{s}_u^{th} = \max \left(\frac{s_{cu}^{th}(\sigma^2 + P_{SI})}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i s_{cu}^{th}) P_b}, \frac{s_{iu}^{th}(\sigma^2 + P_{SI})}{(\alpha_i - (\sum_{p \in \mathcal{K}_i^c} \alpha_p + \sum_{j \in \mathcal{K}_h^c} \Xi_{ju} \alpha_j + \Xi_{cu} \alpha_c) s_{iu}^{th}) P_b} \right) \forall i \in \mathcal{K}. \quad (100)$$

$$\tilde{s}_{ku}^{th} = \max \left(\frac{\hat{s}_{k,cu}^{th} \sigma^2}{(\alpha_c - \sum_{i \in \mathcal{K}} \alpha_i \hat{s}_{k,cu}^{th}) P_u}, \frac{\hat{s}_{ku}^{th} \sigma^2}{(\alpha_k - (\sum_{j \in \mathcal{K}, j \neq k} \alpha_j + \Xi_{cu} \alpha_c) \hat{s}_{ku}^{th}) P_u} \right). \quad (103)$$

considered as $\{0, 0, 0\}$. Further, the other parameter values are tabulated in Table II. Note that all the analytical expressions and simulation results are applicable for any arbitrary value of these parameters.

TABLE II: Simulation Parameters

Parameter	Value	Parameter	Value
$\xi_{kc}^{\max} \forall k \in \mathcal{K}$	10^{-5}	b	3dB [53]
$r_c = r_1 = r_2$	150	P_{SI}	6 dB
$\kappa_c = \kappa_{kp} \forall k \in \mathcal{K}$	75	σ^2	-100 dBm
$\xi_{kp}^{\max} \forall k \in \mathcal{K}$	10^{-5}	β_0	-30dB

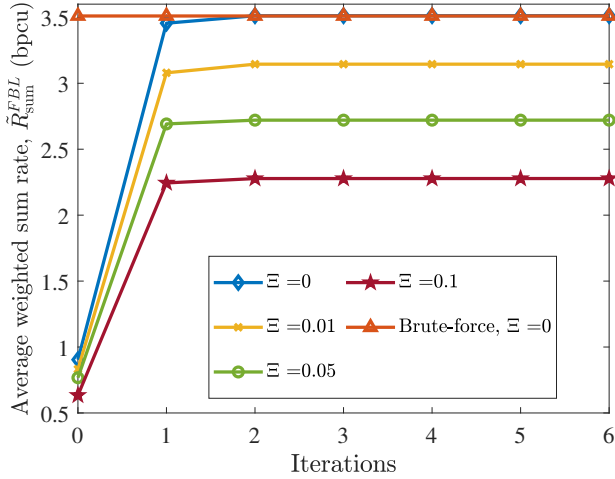


Fig. 2: Convergence

Fig. 2 highlights the convergence behavior of the BCD method. It is obtained using FBL messaging for RIS-FD-UAV-aided transmission mode with the total number of elements $N = 64$ at RIS while the transmit power at BS is considered as $P_b = -5$ dBm. It can be seen that, for the perfect SIC ($\Xi = 0$), the obtained solution converges in a very less number of iterations and achieves a performance closer to the optimal solution which is obtained using the brute-force method with $\Xi = 0$. For example, $\tilde{R}_{\text{sum}}^{FBL} \approx 3.5$ after 3^{rd} iteration for both the methods, the proposed solution, and brute-force, when $\Xi = 0$. Thus, for the considered set of parameters, the proposed method provides a solution that converges after three iterations. Further, it can be seen that the obtained solution also converges for the imperfect case i.e. $\Xi \neq 0$, although due to increasing interference/noise, there is a severe degradation in

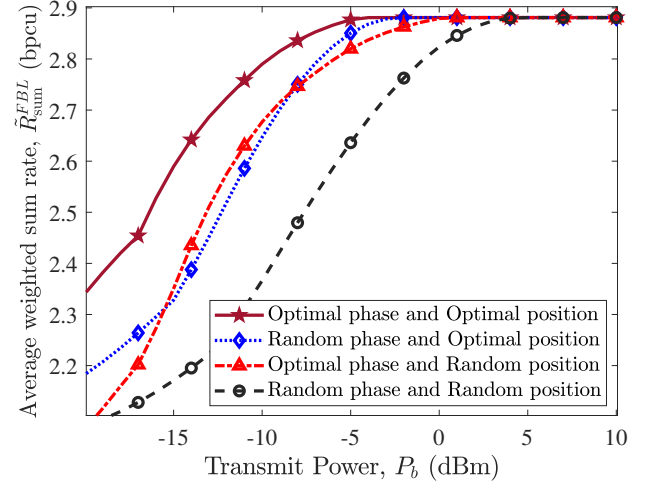


Fig. 3: Average weighted sum rate ($\tilde{R}_{\text{sum}}^{FBL}$) vs transmit power

the maximum achievable $\tilde{R}_{\text{sum}}^{FBL}$ is observed which is caused by imperfections in SIC. For example, with the increase in Ξ from $0 \rightarrow 0.01 \rightarrow 0.05$, one can note the degradation of $3.5 \rightarrow 3.2 \rightarrow 2.75$ in $\tilde{R}_{\text{sum}}^{FBL}$.

Fig. 3 presents the comparison of the average weighted sum rate ($\tilde{R}_{\text{sum}}^{FBL}$) obtained in RIS-FD-UAV-aided transmission mode with FBL messaging for different combinations of phase shift matrix and position (optimal and random). It is obtained by plotting $\tilde{R}_{\text{sum}}^{FBL}$ w.r.t. transmit power (P_b) at BS with $N = 128$ and $\Xi = 0.05$ for four different scenarios: 1) optimal phase and optimal position, 2) optimal phase and random position, 3) random phase, and optimal position, and, 4) random phase and random position. It can be observed that with an increase in transmit power at BS, $\tilde{R}_{\text{sum}}^{FBL}$ also increases. In addition to this, we also observe that the optimal choice of phase shift matrix and position of RIS elements results in the best performance. On the other hand, optimizing any one of either phase matrix or position results in better $\tilde{R}_{\text{sum}}^{FBL}$ than that of randomly choosing both phase and position of RIS elements.

Fig. 4 depicts the plot of the average weighted sum rate for each transmission mode ($\tilde{R}_{u,\text{sum}}^{FBL}$, $\tilde{R}_{I,\text{sum}}^{FBL}$ and $\tilde{R}_{\text{sum}}^{FBL}$) versus the number of RIS elements (N) with transmit power $P_b = -10$ dBm for different values SIC error Ξ . As expected, $\tilde{R}_{u,\text{sum}}^{FBL}$ is independent of N where as $\tilde{R}_{I,\text{sum}}^{FBL}$ increases with an increase in N for both perfect and imperfect SIC. It can be

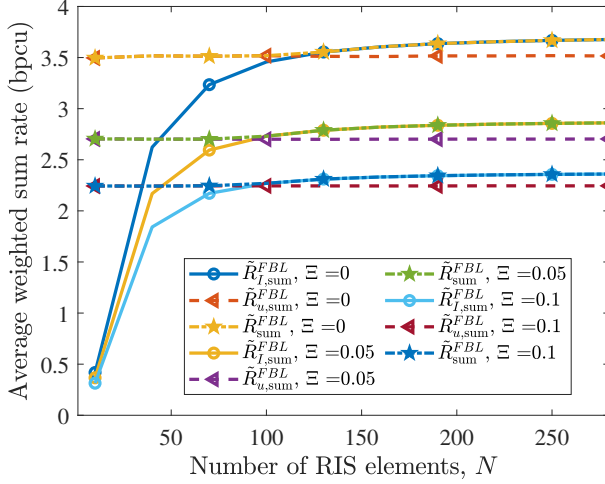


Fig. 4: Average weighted sum rate vs number of RIS elements

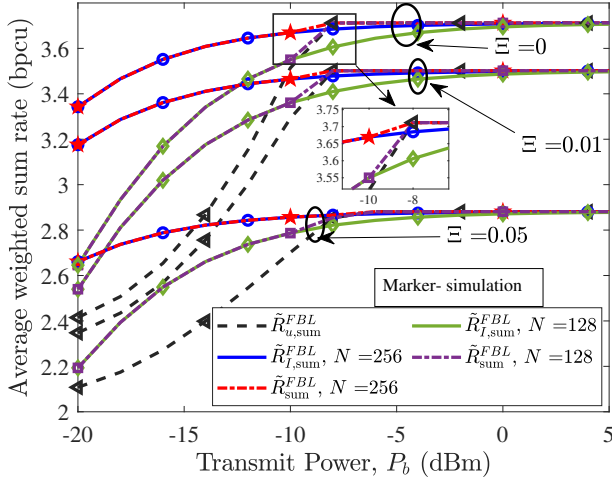


Fig. 5: Average weighted sum rate vs transmit power (P_b)

seen that for the considered parameters, initially (for lower N) UAV-aided mode outperforms the RIS-aided mode due to DF protocol, although after a certain N , the RIS-aided mode dominates the UAV-aided mode. Further, it can also be observed that due to selection combining at each user, the $\tilde{R}_{\text{sum}}^{\text{FBL}}$ is equal to $\max(\tilde{R}_{I,\text{sum}}^{\text{FBL}}, \tilde{R}_{u,\text{sum}}^{\text{FBL}})$, hence providing the best performance among three modes. Moreover, the imperfections in SIC result in a decrease in the weighted sum rate with each mode as compared to that of perfect SIC. We also observe that imperfection in SIC leads to an early saturation in $\tilde{R}_{I,\text{sum}}^{\text{FBL}}$ for the higher values of N . This is because with increasing N , both the desired signal power as well as interference increase and, for higher values of N , an imperfection in SIC results in early saturation.

Fig. 5 depicts the average weighted sum rate of the considered network using different transmission modes with FBL messaging ($\tilde{R}_{u,\text{sum}}^{\text{FBL}}$, $\tilde{R}_{I,\text{sum}}^{\text{FBL}}$ and $\tilde{R}_{\text{sum}}^{\text{FBL}}$) w.r.t. transmit power at BS (P_b) for $N = \{128, 256\}$. It can be observed that the

performance of each user improves with an increase in P_b for each transmission mode. However, due to the imperfection in SIC, there is a significant degradation in the achievable rate. Also, the hybrid RIS-FD-UAV mode outperforms the other two modes throughout the range of P_b and can achieve desired QoS in an RSMA-based wireless network because of SC at each user. Further, it can also be seen that, when $N = 128$, $\tilde{R}_{I,\text{sum}}^{\text{FBL}} > \tilde{R}_{\text{sum}}^{\text{FBL}}$ for $P_b < -7\text{dBm}$, $\tilde{R}_{I,\text{sum}}^{\text{FBL}} < \tilde{R}_{\text{sum}}^{\text{FBL}}$ for $-7\text{dBm} < P_b < 0\text{dBm}$, and $\tilde{R}_{I,\text{sum}}^{\text{FBL}} \approx \tilde{R}_{\text{sum}}^{\text{FBL}}$ for $P_b > 0\text{dBm}$. However, for $N = 256$, RIS-aided mode either outperforms or provides similar performance to that of UAV-aided mode. This indicates that the RIS-aided mode and hybrid RIS-FD-UAV-aided mode provide similar performance when the number of elements at RIS is sufficiently large.

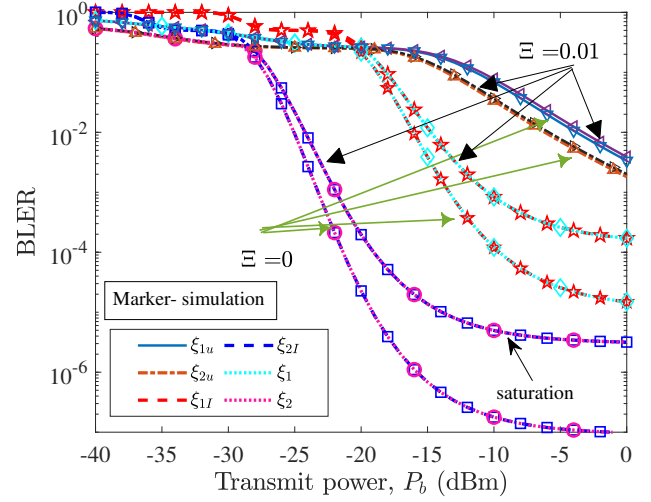


Fig. 6: Average BLER vs transmit power (P_b dBm)

Fig 6 depicts the variation of achievable average BLER at each user using three different mode of transmission (ξ_k, ξ_{ku} and $\xi_{kI} \forall k \in \{1, 2\}$) w.r.t. transmit power (P_b) used by BS for different SIC error (Ξ). One can observe the improvement in BLER with an increase in P_b . The simulation result has

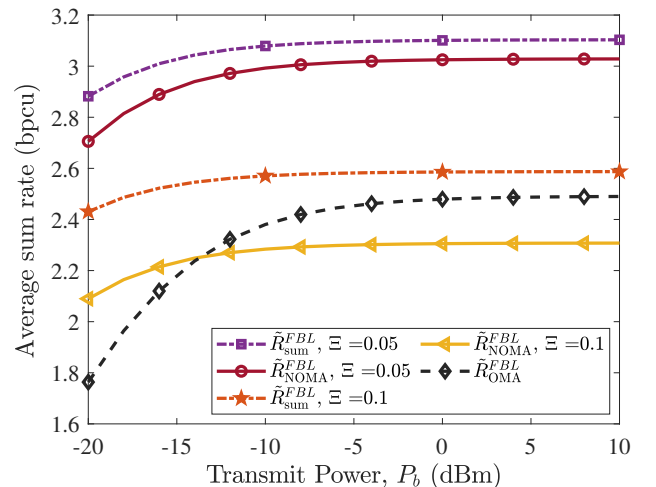


Fig. 7: RSMA vs NOMA vs OMA

proximity to the BLER obtained through derived closed-form expressions in (51), (72), and (86). It can be observed that UAV-aided dominates the performance at low power whereas RIS-aided performance is better than UAV-aided at higher values of P_b . Also, due to selection combining at each user in hybrid mode, the $\xi_k = \min(\xi_{kI}, \xi_{ku})$, hence providing the best performance among three modes throughout the range of P_b . Further, it can be seen that the ξ_k and $\xi_{kI} \forall k \in \{1, 2\}$ saturates at higher power. The reason for this behaviour is that the increase in transmit power increases both the desired signal power as well as the interference power, hence limiting the performance. Moreover, the impact of SIC error is also demonstrated in Fig. 6. It can be seen that the imperfection SIC deteriorates the achievable BLER at each user for all three transmission modes. Also, with an increase in imperfection, the BLER achieves an early saturation ($\xi_{2I} = \xi_2 \approx 10^{-5}$ when $\Xi = 0.01$ after $P_b > -10$ dBm). The reason for this behaviour is that the SIC error leads to additional interference, causing an early saturation to the achievable BLER.

Next, we compare the performance of RSMA with that of NOMA and OMA counterparts by plotting the average weighted sum rate ($\tilde{R}_{\text{sum}}^{FBL}$) for hybrid RIS-FD-UAV transmission mode with imperfect SIC in Fig. 7. Note that $\tilde{R}_{\text{sum}}^{FBL}$ is obtained by considering only private messages with $a_1 = 0.9$, $a_2 = 0.1$ and applying SIC only at GU-2, whereas $\tilde{R}_{\text{OMA}}^{FBL}$ is obtained using equal power distribution for each user and perfect orthogonality among carriers. Clearly, RSMA outperforms the NOMA, as well as the OMA, and provides a higher $\tilde{R}_{\text{sum}}^{FBL}$ for the given set of parameters. Note that the cross over between $\tilde{R}_{\text{OMA}}^{FBL}$ and $\tilde{R}_{\text{NOMA}}^{FBL}$ with $\Xi = 0.1$ is due to the fact that $\tilde{R}_{\text{NOMA}}^{FBL}$ decreases with increase in Ξ whereas $\tilde{R}_{\text{OMA}}^{FBL}$ is independent Ξ , as SIC is not required in OMA signalling.

Fig. 8 depicts the average weighted sum rate of the considered network using different transmission modes with IBL messaging ($\tilde{R}_{u,\text{sum}}^{IBL}$, $\tilde{R}_{I,\text{sum}}^{IBL}$ and $\tilde{R}_{\text{sum}}^{IBL}$) w.r.t. transmit power at BS (P_b) for $N = \{128, 256\}$. It can be observed that the average weighted sum rate at each user improves with an increase in transmit power for each transmission mode. However, due to the imperfection in SIC, there is a significant degradation in the average weighted sum rate. Also, among all three modes, due to SC, performance at each user is maximum using hybrid RIS-FD-UAV mode. Therefore, it can surpass the performance of UAV and RIS-aided modes in achieving desired QoS in an RSMA-based wireless network. Further, it can also be seen that, when $N = 128$, $\tilde{R}_{I,\text{sum}}^{IBL} > \tilde{R}_{\text{sum}}^{IBL}$ for $P_b < -8$ dBm, $\tilde{R}_{I,\text{sum}}^{IBL} < \tilde{R}_{\text{sum}}^{IBL}$ for -8 dBm $< P_b < -3$ dBm, and $\tilde{R}_{I,\text{sum}}^{IBL} \approx \tilde{R}_{\text{sum}}^{IBL}$ for $P_b > -3$ dBm. However, for $N = 256$, RIS-aided mode either outperforms or provides similar performance to that of UAV-aided mode. This indicates that the RIS-aided mode and hybrid RIS-FD-UAV-aided mode provide similar performance when the number of elements at RIS are sufficiently large.

Fig. 9 compares the outage probability at each user with perfect and imperfect scenarios using different transmission modes w.r.t. transmit power at BS (P_b) for $N = 20$. It can be observed that with an increase in P_b , the outage probability at each user decreases for each case. However,

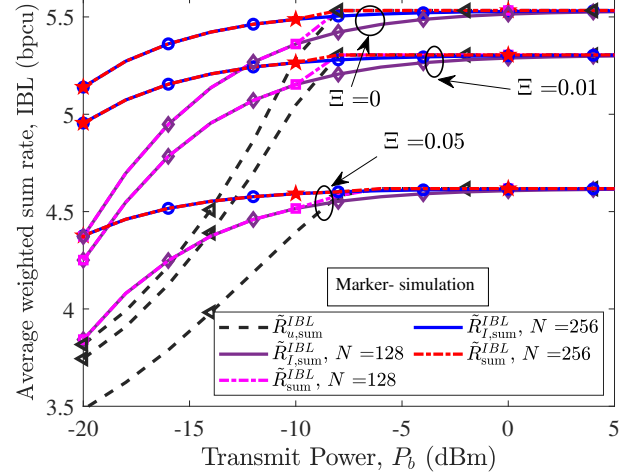


Fig. 8: Average sum rate with IBL vs transmit power (P_b)

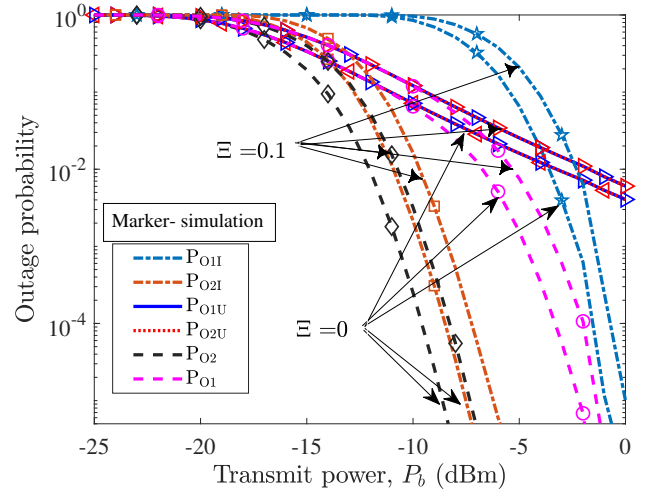


Fig. 9: Outage probability vs transmit power (P_b)

due to the SIC error, there is a significant degradation in the outage performance of each user. Further, for lower P_b , UAV-aided transmission outperforms RIS-aided transmission due to the use of DF relaying. However, for increasing P_b , the performance of the RIS-aided mode significantly improves as compared to the UAV-aided mode due to the presence of the number of reflecting elements. Moreover, for both scenarios, it can also be seen that the hybrid RIS-FD-UAV mode outperforms the rest two modes which is evident from the fact that it selects the best among the two modes for decoding the desired information. Fig. 10 compares the performance of the network considering IBL and FBL transmission schemes. It is obtained by plotting achievable weighted sum rates $\tilde{R}_{\text{sum}}^{FBL}$ and $\tilde{R}_{\text{sum}}^{IBL}$ w.r.t. blocklength $r_c = r_{p1} = r_{p2}$ with $P_b = -10$ dBm. As expected the performance in the case of IBL transmission is better than that of the FBL transmission case. However, one can also observe that the $\tilde{R}_{\text{sum}}^{FBL}$ approaches $\tilde{R}_{\text{sum}}^{IBL}$ as blocklength increases. Thus, we can conclude that $\tilde{R}_{\text{sum}}^{FBL} \approx \tilde{R}_{\text{sum}}^{IBL}$ when blocklength is very large or in other words approaches

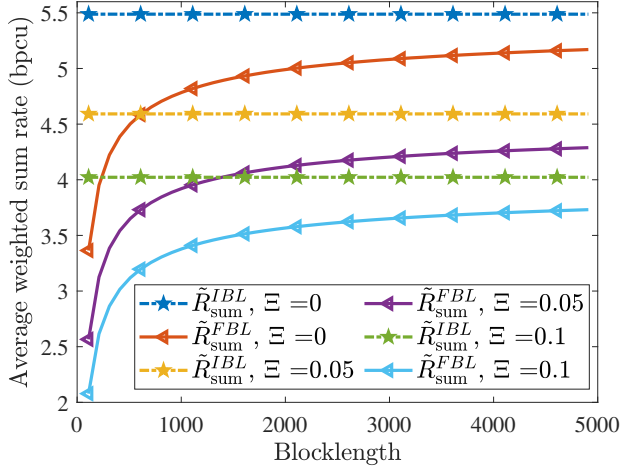


Fig. 10: IBL and FBL vs blocklength

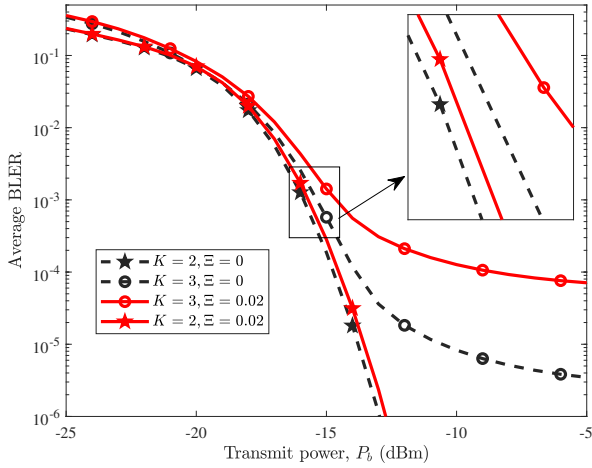


Fig. 11: Average BLER

infinity. Moreover, the SIC imperfection results in a significant decrease in sum rate performance for both the messaging as compared to that of perfect SIC.

Fig. 11 presents the plot of average BLER (ξ_{avg}) obtained using $\xi_{avg} = (1/K) \sum_{k \in \mathcal{K}} \xi_k$ versus the transmit power (P_b) from the BS for $K = \{2, 3\}$ and $\Xi = \{0, 0.02\}$ with $N = 40$, $r_c = r_{1p} = r_{2p} = 500$, and coordinate of third user is set as $\{30, 0, 0\}$. It can be observed that ξ_{avg} decreases with an increase in P_b for each case because of the increase in SINR at each user. Also, the increase in Ξ leads to an increase in ξ_{avg} . Moreover, ξ_{avg} also increases with an increase in K . The main reason for this behaviour is the fact that for a fixed P_b , the net SINR at each user decreases due to a decrease in the net allocated power.

VII. CONCLUSIONS

In this work, we introduced the use of a hybrid RIS-FD-UAV-aided relaying in a multiuser communication network using RSMA signalling for both IBL and FBL messaging. We adopted the BCD method to solve the primal problem of

weighted sum rate maximization by breaking it into multiple sub-problems. Thereafter, a BCD-based algorithm was proposed that jointly optimizes these parameters using a heuristic approach for optimum power allocation, a Riemannian conjugate gradient-based algorithm to get the optimal phase shift at the RIS, and an iterative algorithm to obtain the optimal UAV/RIS position. In addition to these parameters, it also optimized the distribution of the common rate among all the users. The proposed method provides near-optimal solutions but with much lower complexity as compared to the conventional methods. Furthermore, we also analyzed the performance of the considered framework by deriving the closed-form expressions of BLER, outage probability, and average weighted sum rate and highlighted the importance of the required number of RIS elements in order to dominate the achievable performance as compared to UAV-aided mode. Next, the effectiveness of the proposed algorithms was validated through exhaustive simulation results, and we found that the hybrid RIS-FD-UAV transmission mode achieved the best performance as compared to the other two conventional methods for both IBL and FBL messaging schemes. Additionally, we discussed the impact of reflecting elements, the number of users, and available transmit power on the system's performance.

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